

Macro-eco-evolutionary perspectives on the Cenozoic history of South American Mammals

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Foreword and Acknowledgements

Foreword

Acknowledgements

1 General Introduction

1.1 The unevenness of biodiversity: corner stone for the large-scale study of life

The astonishing diversity of life on Earth has fascinated scientists for centuries. A study dating back to 15 years ago indeed estimated that around 8.7 million eukaryotic species are currently coexisting on our planet ([Mora *et al.* 2011](#)). This number would probably be significantly increased by accounting for the substantial improvements that have occurred in the field of systematics over the recent years, and expanded by orders of magnitude if prokaryotic species were incorporated. Yet, someone travelling to the Sahara desert would probably be less amazed by the surrounding diversity of life than someone travelling to the middle of the Amazonia, one of the most species-rich regions in the world ([Raven *et al.* 2020](#)). Another vertiginous data is that around 99.9% of the species that ever existed since the dawn of life on Earth, some four billion years ago, has gone extinct ([Raup 1986](#)). It therefore stands out that, far from a fixed and smooth entity, biodiversity, here understood as species richness, presents a myriad of heterogeneities, that stimulated a vast quantity of research works in which this thesis happens to be nested.

1.1.1 The heterogeneous organisation of life across space

For more than two centuries, it has been recognised that species are not randomly distributed on Earth. In his pioneering “*Essai sur la géographie des plantes*”, regarded as foundational for the science of Biogeography (see [Glossary](#)), Alexander von Humboldt emphasised the spatial nature of plant taxonomic assemblages ([Humboldt & Bonpland 1807](#)). Later in the 19th century, the scientific community acquired a uniquely integrated perspective on biodiversity fed by years of field observations, giving to the question of the spatial organisation of life on Earth a new dimension. Consequently, it has been proposed to subdivide the world into spatial units sharing similar fauna/flora, sometimes called ‘Homozoic Belts’, ‘Regions’, ‘Realms’, ‘Provinces’, ‘Areas’ or ‘*Gebiet*’ (German word for territory, area) ([Burkill 1943](#)). The history will mainly remember the efforts of Alfred R. Wallace in that respect, who relied on the subdivision made by Philip L. Sclater in 1874 based on mammals, as well as his own expertise, to establish the famous contouring of the Earth’s ‘Zoogeographic Realms’ (Figure 1.1), two years after ([Wallace 1876](#)). Further efforts towards incorporating fossil evidence to better constrain the aforementioned subdivisions have to be acknowledged, notably implused by Richard Lydekker ([Lydekker 1896](#)). Wallace’s subdivision has maintained in the field since its publication, and only received minor updates in the last decade ([Holt *et al.* 2013](#)), illustrating the degree of knowledge advancement of the English naturalist, in a context where Charles Darwin’s famous Theory of Evolution – that, actually, Wallace co-discovered in a totally independent manner, making such a theory one of the most striking examples of thought convergence in the history of sciences ([Berry & Browne 2008](#)) – still had a long way to go before being formally accepted by the broad scientific community.

Beyond its spatial organisation into units of shared taxonomic composition, biodiversity also ex-



Figure 1.1: Biogeographic map based on faunal similarities within animal assemblages across space ('Zoogeographic Realms') proposed by Wallace (1876).

hibits a striking latitudinal organisation, in which species richness increases with decreasing latitude, known as the Latitudinal Biodiversity Gradient (Figure 1.2). Such a pattern was already recognised by Humboldt. According to him, low-latitude plant tropical assemblages would be essentially made up with a large number of different species per unit of area, each having a relatively low density ("isolated and sparse"). Conversely, higher-latitude temperate or cold assemblages would be characterised by a substantially lower species richness per unit of area, with a single dominant species being over-represented ("excluding [...] heterogeneous species") (Humboldt & Bonpland 1807). Half a century later, in his cornerstone book "*The Origin of Species*", Darwin dedicated two chapters to the spatial distribution of species, recognising that many groups harbour a greater diversity in the tropics and seeking for potential mechanisms underlying this pattern (Darwin 1859). Today, though notable exceptions exist (*e.g.*, pinnipeds, lagomorphs or several kinds of fungi), a latitudinal diversity gradient has been recognised among most taxonomic group (*e.g.*, plants, insects, molluscs, vertebrates) (Hillebrand 2004), as depicted on Figure 1.2 for three vertebrate classes.

Hence, biodiversity is largely heterogeneous across space, raising questions that have been thoroughly investigated in the past, as to 'Why are there more species in the tropics than in the poles?' or, more generally, 'What constraints the distribution of species on our planet?'

1.1.2 From peaks to declines: temporally-heterogeneous biodiversity

The aforementioned complex spatial arrangement of life is in fact the legacy of hundreds of millions of years over which species have been fashioned by a wealth of forces – of which we intend to provide an overview in Section 1.2.1. The statement that species evolve across time, and that species-level temporal changes eventually scale up to multispecies assemblages making up biodiversity is now commonly accepted, mostly thanks to fossil data.

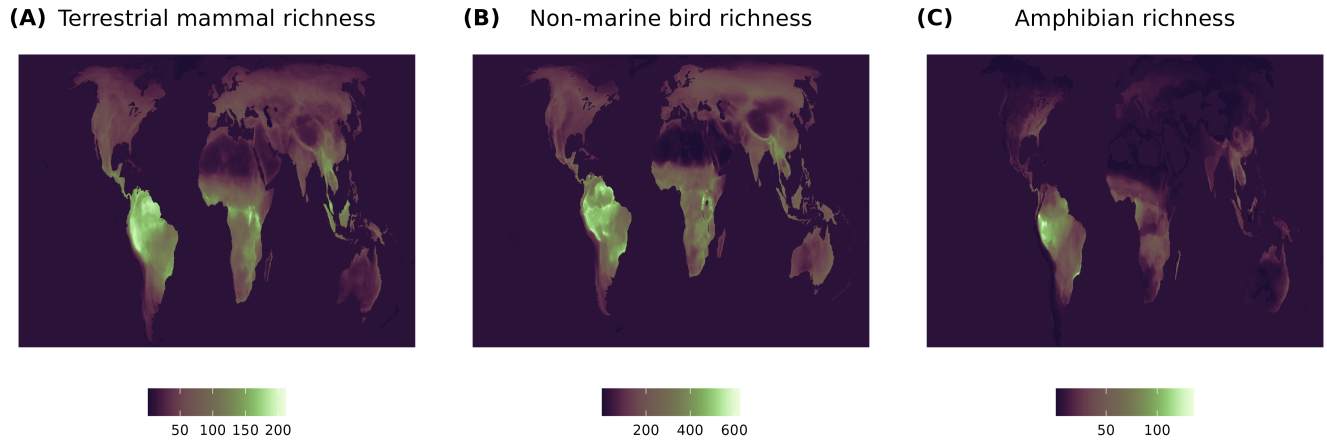


Figure 1.2: Extant land mammal (A), non-marine bird (B) and amphibian (C) species richness at the surface of the globe, expressed in number of species, with a 10 x 10km resolution. Data are from <https://biodiversitymapping.org/>

Being the only source of direct evidence of life in the past, fossils have indeed been of immense value to characterise the temporal dynamics of biodiversity, notably since they irrefutably document the occurrence of extinction in the history of life. It is however interesting to notice that the reasoning that makes us conclude that if, for instance, mammoths are found in the fossil record but not among extant assemblages, it is because they have gone extinct, has not always been as straightforward as it appears to us. Fossils have indeed puzzled eminent figures in the field of biology, such as Jean-Baptiste de Lamarck – to whom we incidentally owe the name ‘biology’. In the early 1800s, Lamarck tentatively formalised a theory where animals would be subject to a complexifying force driving body plans towards higher levels. According to his theory, the (animal) biosphere would thus represent a continuum where species would be unidirectionally connected between each others, which, importantly, leaves no room for extinction – or any sort of species richness variation across time – : every member of Lamarck’s gradual sequence towards increasingly-complex body plans would already exist (Lamarck 1815). However, Lamarck had to cope with the fact that the field of systematic palaeontology was emerging in parallel, notably impelled by his colleagues from the Muséum National d’Histoire Naturelle de Paris, including Georges Cuvier (Servais *et al.* 2012). Plenty of remains of animals trapped inside the rocks but also manifestly absent from the surface of the (known) world were thus unearthed and described, challenging Lamarck’s model. In order to make sense of this apparent inadequacy, Lamarck formulated a famous *ad-hoc* hypothesis: the apparent absence of extant representatives of mammoths does not mean that mammoths have gone extinct. If they indeed locally disappeared from the places where they left fossil remains, they must have found refugia in a region of the globe unexplored by the occidental civilisation.

Markedly opposed to Lamarck’s theory of animal organisation following increasingly complex body plans, Cuvier was maybe one of the earliest scientists to propose that species assemblages could change across time, although the theory that he developed (‘catastrophism’) was largely invalidated later in the history. His rich fieldwork expertise indeed led him to postulate that the history of the globe has been punctuated by abrupt ‘revolution’ events, which can be viewed as (abiotic) changes occurring within a given environment, in turn causing changes in the nature of the species present within this changing environment (Cuvier 1840). Two decades later, John Phillips, an English palaeontologist notably known for giving Phanerozoic eras their name (Palaeozoic, Mesozoic and Cenozoic) compiled

for the first time fossil occurrences of marine taxa from the British lithological record, that he used to produce the first continuous diversity-through-time curve, spanning the entire Phanerozoic (Phillips 1860). His work allowed him to discriminate what he called the Palaeozoic, Mesozoic and Caenozoic ‘lives’. These three ‘lives’ were separated by vast depressions, corresponding to the Permo–Triassic and Cretaceous–Palaeogene boundaries (Figure 1.3 A), that will be later identified as two amongst the five most significant extinction episodes that marked the history of life, also known as mass extinction crises (see Section 1.2.1). Phillips’ cornerstone compilation thus, for the first time, quantitatively demonstrated the heterogeneity of species richness across geological times.

A century later, the rise of quantitative palaeobiology (see Section 1.1.4) paved the way towards more accurate reconstructions of biodiversity across time. The publication, in the early 1950s, of the first edition of the *Treatise of Invertebrate Paleontology* provided an access to a vast quantity of marine invertebrate fossil data, that could be further used for such an achievement. In that framework, Norman Newell reconstructed diversity curves for about 16 different groups (Newell 1952). Again, the reconstructed curves displayed time-varying patterns. Newell noticed certain similarities in the reconstructed patterns between groups, that he suggested to be evidence for “common causes” underlying their large-scale evolutionary trends. A few decades later, with a higher-resolution curve, Valentine (1969) (Figure 1.3 B) proposed to “explain diversity trends in terms of ecological hierarchy” (Sepkoski 2012), constituting a significant advance towards integrating ecological and evolutionary processes (see Section 1.2).

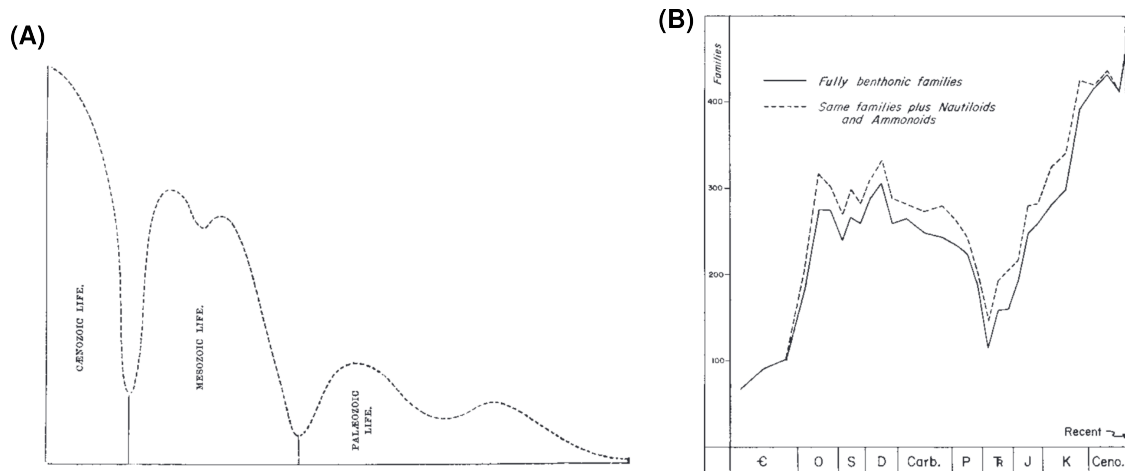


Figure 1.3: Historical fossil-based reconstructions of marine diversity across geological times produced by Phillips (1860), using a marine invertebrate compilation from the British lithological record (A) and by Valentine (1969), using marine invertebrate record from the *Treatise of Invertebrate Paleontology* (B). Note that timescale in (A) has been reverted when compared (B), going back in time from left to right instead of right to left.

However, Valentine’s reconstructions raised some methodological criticism, especially expressed by David M. Raup, one of the leading figures of the ‘palaeobiological revolution’. Raup was indeed concerned about how the fossil record biases may affect Valentine’s reconstruction, importantly stressing that “the increase in the number of marine species across the Phanerozoic may be more apparent than real” (Raup 1972, see Section 1.1.4). Ten years of intense data compilation and methodological labour after, together with J. John Sepkoski Jr. (another major actor of the ‘palaeobiological revolution’) Raup came to the conclusion that marine biodiversity indeed experienced temporal variations, persisting even after accounting for biases inherent to the fossil record (Sepkoski *et al.* 1981). A year after,

Raup & Sepkoski (1982) characterised major temporal features in their diversity curve (Figure 1.4), now known as the ‘Big Five mass extinctions’ (that we will develop in further details in Section 1.2.1).

Nowadays, it makes no doubt that biodiversity is temporally heterogeneous. Clades originate, thrive and then possibly decline or even go extinct, sometimes fluctuating in species number in between. A core aim of macroevolution, in which this doctoral work is nested, is thus to elucidate, given a clade, how species diversity changes across time, and the factors underlying these changes (Section 1.2.1).

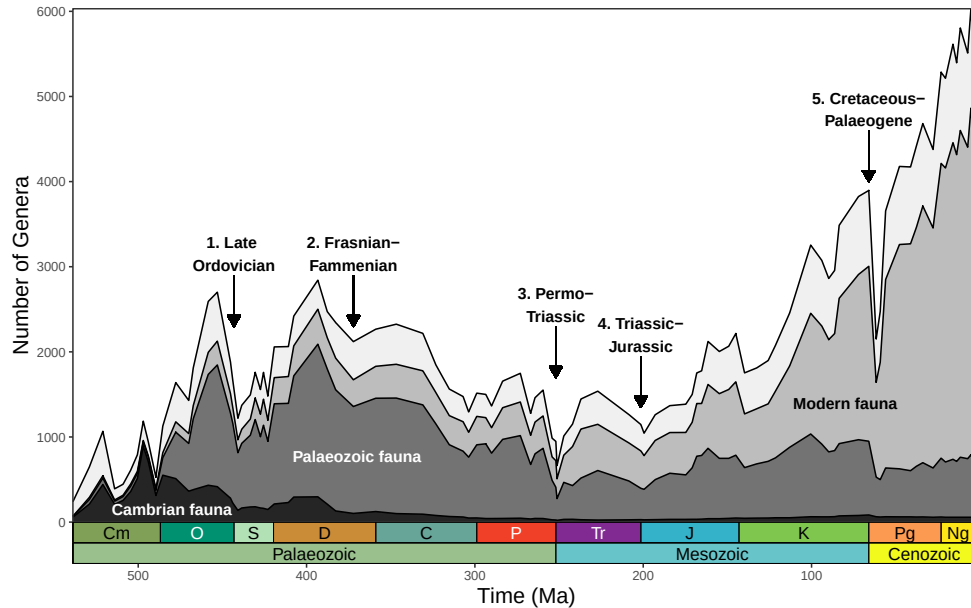


Figure 1.4: Temporal variations in marine animal biodiversity across the Phanerozoic. Colours indicate the three evolutionary faunas discriminated by Sepkoski (1984). Taxa that could not be assigned to any fauna are included in the clearest band. Arrows indicate the ‘Big Five’ mass extinctions, as defined by Raup & Sepkoski (1982). Data are from the genus-level compendium released by Sepkoski (2002), binned across geological stages, accessed and displayed thanks to the R package `sepkoski` (Jones 2022). Geological timescale was added thanks to the `deeptime` R package (Gearty 2023).

1.1.3 Inter-clade heterogeneity

Another facet of the heterogeneity of biodiversity comes from the fact that some groups harbour an immensely larger species richness than others. This striking feature has intrigued evolutionary biologists since the times of Darwin. In a 1879 letter to Joseph Hooker, the renowned white-bearded English naturalist indeed famously termed the astonishing diversity of flowering plants (Angiosperma) and their apparent rapid burst in the fossil record as an ‘abominable mystery’. At present, with more than 300,000 described species, angiosperms encompass nearly 90% of the extant land plant diversity.

Over the past few decades, the rise of phylogenetics (Section 1.3.1) permitted to better constrain evolutionary relationships between lineages, thus highlighting inter-clade diversity disequilibria (Figure 1.5). Understanding what drives striking diversity disequilibria between groups, even sometimes closely related, is a central aim of macroevolutionary research.

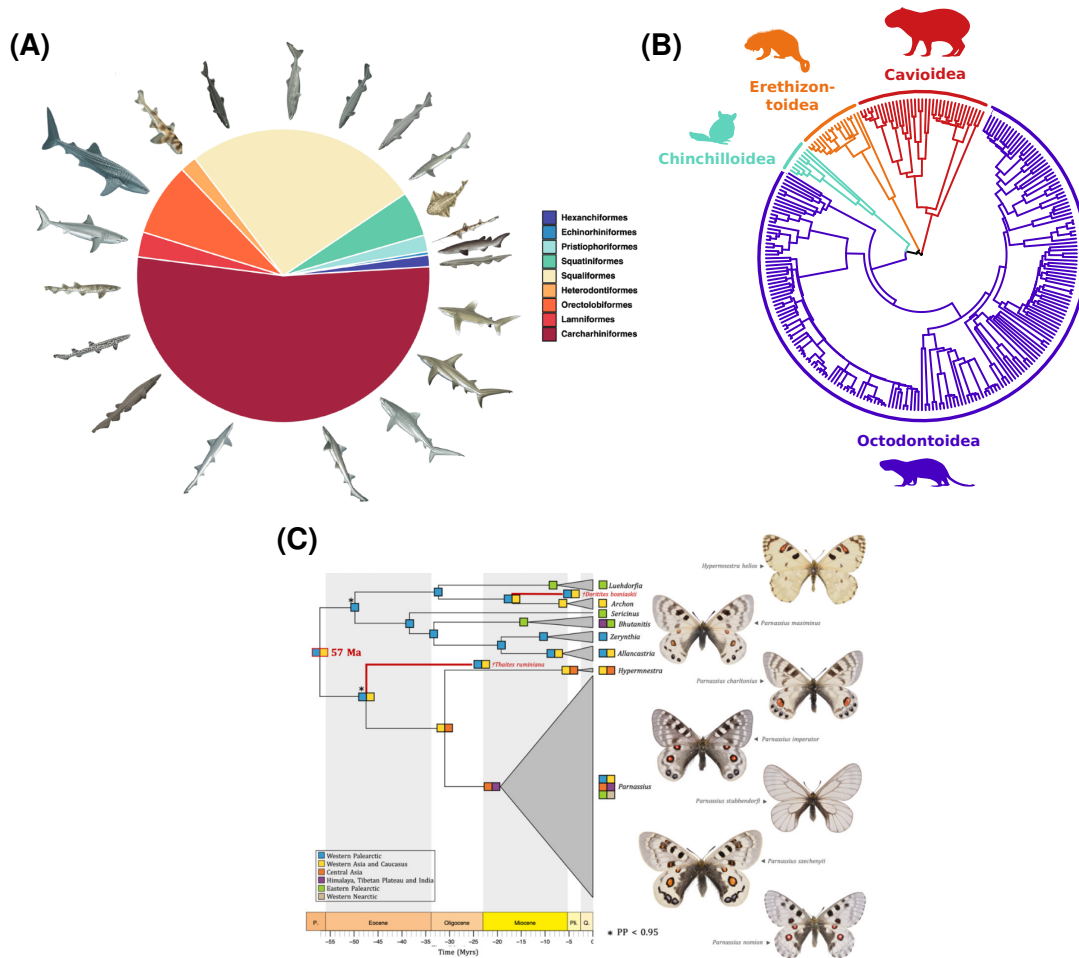


Figure 1.5: Examples of diversity disequilibria in the animal tree of life. **(A)** >50% of the extant shark (Neoselachii) diversity is comprised within the order Carcharhiniformes, whereas other orders (*e.g.*, Echinorhiniformes, Hexanchiformes) are comparatively way less diversified (figure reproduced with permission from Marion *et al.* (2024)). **(B)** ~80% of the extant caviomorph (Rodentia: Hystricognathi) diversity is comprised within the superfamily Octodontoidea, whereas the Chinchilloidea (see *Chapter 2*) only comprises 8 living species. Phylogeny is from VerLife, and silhouettes are from Phylopic (Credits : Michael Tripoli and Gabriela Palomo Muñoz). **(C)** ~90% of the extant diversity of Parnassiinae (Lepidoptera: Papilionidae) is comprised within the genus *Parnassius* (figure reproduced with permission from Hévin & Condamine (2024)).

1.1.4 Methodological biases arising when studying biodiversity

Biodiversity shortfalls

Several caveats apply to the study of biodiversity patterns ‘from a distance’, the baseline of the macroecological perspective (Diniz-Filho 2023). Some of them are grouped into what is known as the ‘biodiversity shortfalls’, representing inherent limitations related to missing information in biodiversity

data, thereby inevitably hampering accurate understanding of the modelled eco-evolutionary process. Three biodiversity shortfalls are usually considered, though Hortal *et al.* (2015) in fact more finely discriminated seven – that will not be extensively covered here. These three have been named after three renowned figures of ecology, evolution and systematics.

The *Linnean shortfall* (named after Carl von Linné, the Swedish father of systematics) reflects the α -taxonomic uncertainty, that is, the degree of certainty that one may have, after sampling two individuals at random in a community, to tell whether they belong to the same species or to two distinct species. Materialisations of the Linnean shortfall are, for instance, cryptic diversity (under-estimation of the number of species within a clade due to a lack of data and/or morphological similarities), or ‘taxonomic splitting’ (over-estimation of the number of species within a clade, often arising from the use of morphological criteria; it therefore particularly applies to the fossil record). Next, the *Wallacean shortfall* (named after Alfred R. Wallace, to whom this thesis will refer several times) refers to the knowledge gaps in species distribution, *i.e.*, the fact that a species may be somewhere where it has not been observed; thus this place in particular would be excluded from upstream range map reconstructions of that species (sort of type-1 error applied to biodiversity data). Last, the *Darwinian shortfall* (named after Charles Darwin, also mentioned several times across this thesis) refers to the lack of knowledge of the evolutionary relationships between species.

Additionally, the study of biodiversity has long been affected by cultural and/or geopolitical constraints, either limiting the broad accessibility of published knowledge (*e.g.*, written in non-latin alphabets) or the access to specimens and/or sampling localities.

Propagation of the caveats of the fossil record when studying biodiversity through time

“A single answer to the question of whether the fossil record is driven by macroevolution or megabias is unlikely ever to emerge because of temporal, geographical, and taxonomic variance in the data.” (Benton *et al.* 2011)

Additional biases emerge from reconstructions of biodiversity patterns over long time scales, overlaying on the previously described biodiversity shortfalls. These reflect the inherent incompleteness of the fossil record.

Darwin himself emphasised the imperfection of the fossil record (somewhat understandable given how poor the fossil record was at his time compared to the present-day), comparing the latter to a book written in different dialects with many missing pages, and with only a few lines preserved on the remaining pages (Darwin 1859). This conception of the fossil record, too sparse to be accurately used to reconstruct evolutionary processes (“Darwin’s dilemma”, Sepkoski (2012)) largely spread among evolutionary biologists. Consequently, during most of the 20th century, palaeontology was marginalised from the rest of the evolutionary biology community. Roughly between 1970 and 1985, a massive change of paradigm occurred as some palaeontologists started to focus on large-scale analyses of evolutionary patterns through quantitative analyses of the fossil record, known as the “palaeobiological revolution” (Sepkoski 2012). Increasing attention was dedicated to fossil-related biases, giving birth to historical controversies about the way to “read” through the fossil record, as the famous one that opposed James W. Valentine and David M. Raup (Raup 1972; Valentine 1969). This period thus paved the way towards significant methodological development to infer processes related to the temporal dynamics of biodiversity accounting for several known limitations of the fossil record, yielding unprecedented findings calling many established paradigms on the dynamics of biodiversity into question (*e.g.*, Alroy *et al.* 2001; Barrett *et al.* 2009; Benson *et al.* 2016; Close *et al.* 2020; Smith

& McGowan 2007).

Essentially, fossil-related biases can be grouped into three main classes. The first one naturally relates to the exceptional phenomenon of *fossilisation*, operating a selective transition from stacks of organic matter to permanent imprints trapped inside the rocks (Cook *et al.* 1961). Fossilisation requires a very special array of environmental conditions that will synergetically interact to by-pass the almost inevitable process of organic matter decomposition (*e.g.*, rapid burial, anoxic conditions). Its selective nature has several facets. The *structural selectivity* of fossilisation lies within the fact that, provided the aforementioned environmental conditions are met, mineralised tissues (*e.g.*, teeth, bones) will be more easily preserved than soft tissues, the latter being almost never preserved (except in very exceptional cases, Shields (1998)). Furthermore, the *ecological selectivity* of fossilisation has to do with organisms' ecological preferences towards zones of high fossilisation potential. Consequently, fossilisation is an extremely rare and selective process.

The second class of biases arising from the fossil record is more human-related, as it involves the *sampling* process. Indeed, unearthing a fossil has some parts of contingency. Should a hypothetical excavation be conducted a few metres away from a spot hosting a highly-valuable fossil, such a fossil would probably never be found. That being said, far from being uniformly distributed at the surface of the globe, sampling effort exhibits a striking spatial structure (Smith 2001), being extremely high in some regions (*e.g.*, North America, Europe, Japan, Argentina), and extremely low in others (*e.g.*, tropical non-Andean northern South America, Africa, South East Asia, Siberia, Antarctica, Western Australia) (Figure 1.6). It is interesting to notice that the most species-rich regions at present (*i.e.*, the tropics, Figure 1.2) are often the most fossil-depauperated (Figure 1.6). Consequently, the *known fossil record* (*i.e.*, all the specimens that have been unearthed to date) is only a subsample of the *potential fossil record*, (*i.e.*, all the biological remains that came to fossilise in the history of life), itself a subsample of all the life that ever existed (Benton *et al.* 2011). The spatial structure of the fossil record is an ongoing research avenue, that led to several keystone discoveries (*e.g.*, Antell *et al.* 2024; Close *et al.* 2017; Flannery-Sutherland *et al.* 2022; Reijenga & Close 2025).

For a given taxon, the stochastic character of both sampling and fossilisation may lead to a significant truncation of its temporal persistence. In that case, the oldest and youngest occurrences of this taxon may fail to accurately reflect its actual origination and extinction ages, incorporating a significant bias to any upstream occurrence-based analysis. The importance of this bias raised particular concern for the detection of mass extinction events (Section 1.2.1), since many of the lineages actually going extinct during such events would be expected to disappear from the fossil record prior to its realisation. These artificially truncated age ranges would therefore tend to highlight a gradual extinction pattern starting well before the actual extinction, which goes in contradiction with the abrupt character of mass extinctions (see Glossary). This well-known bias has been termed ‘Signor-Lipps’ effect, from the names of the two authors first describing it (Signor & Lipps 1982).

The third class of biases can be referred to as time-related biases. These comprise the difficulty to replace a fossil in an absolute temporal framework (Behrensmeyer 2026). Indeed, dating a fossil is far from being a trivial task. In the best scenario, nearby absolute anchoring points (*i.e.*, minerals usually from tuffs or volcanic zircons that can be robustly dated with isotopes) allow to constrain the interval when the fossil-bearing beds were deposited (*absolute dating*). However, many deposits lack such temporal anchors, and are thus dated using indirect methods, that will not be detailed here (*relative dating*). Hence, age intervals coming along with a fossil can be large, which may in turn add another layer of uncertainty to upstream biodiversity-through-time reconstructions. Another source of time-related biases is the fact that the probability for a fossil to get eroded, and thus, lost, increases with its age. Consequently, the number of available fossil occurrences is expected to be larger for

younger assemblages, an effect termed ‘Pull of the Recent’ ([Foote 2000](#)).

Thus, studying large-scale biodiversity patterns across time requires embracing many sources of uncertainty.

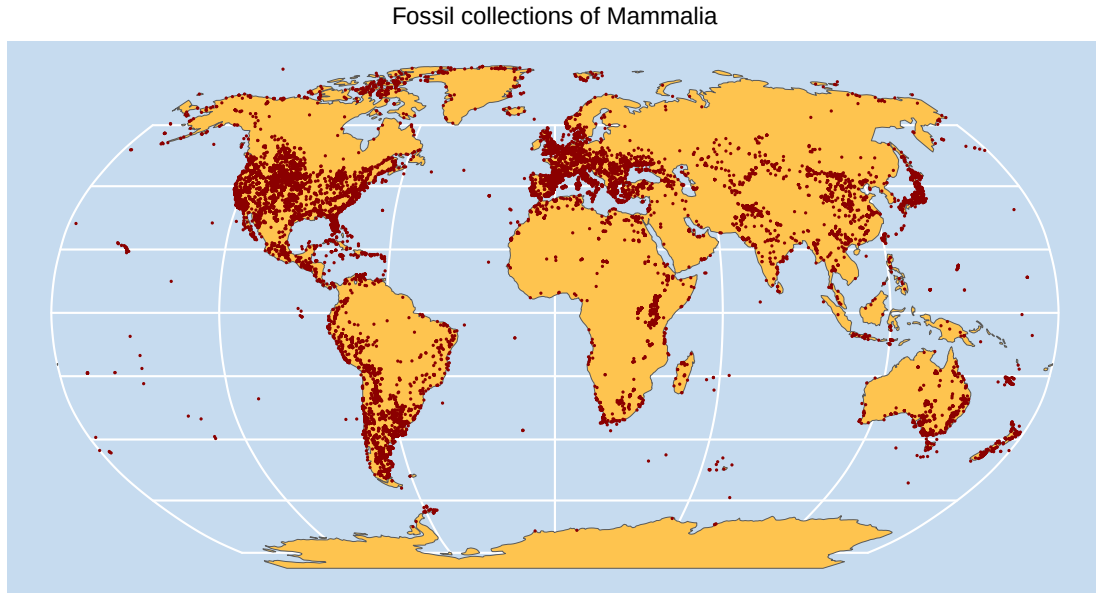


Figure 1.6: The spatial heterogeneity of sampling effort, here exemplified by the global distribution of mammal fossil collections (all geological periods confounded). Data from [The Paleobiology Database](#).

1.2 Disentangling biodiversity patterns across space and time: the macro-eco-evolutionary perspective

1.2.1 Tempo and mode of macroevolution

Macroevolution is usually understood as the study of evolutionary patterns occurring above the species level ([Hautmann 2020](#); [Saupe & Myers 2021](#)). It has been classically approached under two non-mutually exclusive lenses ([Stanley 1979](#)), either focusing on (1) the diversification process ([Glossary](#)), or (2) the evolutionary dynamics of phenotypes – as an echo to the term ‘diversification’, some authors refer to the process underlying morphological diversity as ‘disparification’ (*e.g.*, [Davis *et al.* 2016](#); [Nürk *et al.* 2019](#)). Unfortunately, despite their undeniable appeal, questions related to the dynamics of morphological evolution fall a bit outside the scope of this thesis. Thus, in this manuscript, emphasis will be placed on clade diversification (see [Glossary](#)), that is, the balance between the speciation and the extinction processes. The heterogeneity of biodiversity can be indeed viewed as the result from variations in the pace at which lineages originate and disappear, namely, extinction and speciation rates. Deciphering when (*tempo*), how and why (*mode*) these rates vary across space, time and clades is a core aim of macroevolution ([Saupe & Myers 2021](#)).

Tempo of Macroevolution: when do rates change across time?

As seen in the previous section, the history of life has been punctuated by important fluctuations in diversity, possibly heterogeneous across lineages and geographical areas. Marked increases in diversity, either generated by an increase in speciation rate or a decrease in extinction rate (or a combination of the two), known as evolutionary radiations, have long been recognised (*e.g.*, [Losos 2009](#); [Simões *et al.* 2016](#)). The most popular examples of evolutionary radiations are probably adaptive radiations, in which the acquisition of a ‘key innovation’ spurs the rapid diversification of a clade accompanied by eco-morphological divergence ([Gavrilets & Losos 2009](#); [Glor 2010](#); [Simpson 1953](#)). This concept applies to a wide range of groups across the tree of Life. One may for instance refer to the angiosperms, of which the flower (along with other traits related to physiology, like the pace of living) is thought to have promoted their Late Cretaceous rapid diversification, Darwin’s ‘abominable mystery’ ([Benton *et al.* 2022](#); [Sauquet *et al.* 2017](#)). Bats, the second most species-rich mammalian order, also stand out as an interesting example of adaptive radiation, their diversity being tightly related to the emergence of echolocation ([Jones & Teeling 2006](#)). Last, the radiation of hummingbirds (Aves: Trochilidae), accounting for nearly 400 species at present, can also be viewed as an example of adaptive radiation, enhanced by the acquisition of hovering flight and nectarivory ([Mayr 2004](#)). However, radiations are not always adaptive, and several works of the last few decades have called for terminological clarifications in that respect ([Glor 2010](#); [León-Tapia *et al.* 2026](#); [Simões *et al.* 2016](#); [Tu *et al.* 2025](#)). The extant tree of life indeed offers striking examples of huge and rapid lineage accumulation that has not been accompanied by ecological divergence. This is for instance the case of Sigmodontinae (Rodentia: Cricetidae), the second most diversified rodent clade, and the most diversified mammalian group in the American tropics ([León-Tapia *et al.* 2026](#); [Maestri *et al.* 2017](#)).

Furthermore, periods of diversity loss have long been recognised in the fossil record. These are well exemplified by the abrupt disappearance of trilobites or non-avian dinosaurs, respectively at the Permo–Triassic and the Cretaceous–Palaeogene boundaries. However, the biological relevance of these apparent extinction episodes has been called into question by the arising field of statistical palaeontology in the early 1970s ([Sepkoski 2012](#)). Two levels of concerns were expressed: (1) could these apparent extinction episodes in fact only reflect the incompleteness of the fossil record (Section 1.1.4) and (2) could these fluctuations differ from the predictions of a model in which speciation and extinction are purely random processes (*e.g.*, [Gould *et al.* 1977](#); [Raup *et al.* 1973](#); [Raup 1978](#))? After a decade of methodological improvements, it has been established that ‘natural clades’ had features that could not be reproduced with simple non-mechanistic stochastic models ([Sepkoski *et al.* 1981](#); [Stanley *et al.* 1981](#)). Coming along with external sources of validation (*e.g.*, the characterisation of an asteroid impact using Iridium anomalies at the precise time when dinosaurs disappeared from the fossil record ([Alvarez *et al.* 1980](#))), five time frames with significantly elevated extinction levels over the Phanerozoic were thus characterised, formalised as the ‘big five mass extinctions’ ([Raup & Sepkoski 1982](#)) (depicted on Figure 1.4). Further research, first facilitated by the release of the high-resolution genus-level compendium of marine animals made by Sepkoski (2002), and then, by collective fossil data compilation efforts (*e.g.*, [The Paleobiology Database](#)) in fact transposed the definition of mass extinctions to other intervals of the Phanerozoic, stressing that these periods could show variable degrees of extinction magnitude ([Bambach 2006](#); [Hoyal Cuthill *et al.* 2020](#)). This thesis will add some elements of discussion regarding the mass extinction status of one of these later-characterised mass extinction intervals, the Eocene–Oligocene Transition (see Chapter 1). That being said, mass extinctions arguably account for a minority of the extinctions that have occurred throughout the history of life ([Quental & Marshall 2013](#)). As stressed by Raup (1986), biodiversity is also affected by a “spectrum of smaller events” occurring at the scale of individual clades, or even lineages. Extinction

is thus not always abrupt: it can also operate gradually and, noteworthy, diversity losses occurring through the history of life may also be driven by decreases in speciation/origination rates (thus, not only by elevated extinction rates). This often overlooked perspective was termed the ‘Entwives effect’ by Quental & Marshall (2013), a metaphor from John R. R. Tolkien’s *Lord of the rings*, in which the Ents, a race of anthropomorphic trees, lost their wives, and thus had no means to have a descent (or to originate, in macroevolutionary terms).

Consequently, there is compelling evidence supporting that diversification rates varied across the history of life, and that these variations may be ruled by different evolutionary regimes (*e.g.*, gradual vs. mass extinction). It is now appropriate to move on to the factor(s) underlying these rate variations.

Mode of Macroevolution: what drives rate variations?

The search for the mechanisms causing temporal changes in diversity – and thus diversification rates – has naturally come along with the first macroevolutionary reconstructions (Section 1.1.2). Noticing similar trends in the diversity curves of independent marine invertebrate clades, Newell (1952) immediately postulated the existence of “common causes” to which these clades must have been subject. These causes would “undoubtedly [be] complex and include both biological and physical factors” (Newell 1952). Nowadays, things have not gone farther that much: biodiversity is still considered as being affected by a mixture of biotic and abiotic factors (Benton 2009). Below is an attempt to produce a brief outline of the cornerstone rules that have been formulated to relate the pace of clade diversification to external drivers.

In 1973, Leigh Van Valen published a groundbreaking study that intrigued the community, starting from himself (Van Valen 1973). He evidenced that the fossil-derived survivorship curves of a wide range of taxa were mostly log-linear. This ‘Law of Constant Extinction’ implies that, at any time point in the evolutionary history of a clade, all the members of this clade are equally-likely to go extinct, irrespective of their age (Figure 1.7 B). This goes against alternative hypotheses instead stating that extinction rates should increase with taxon age (namely, age-dependent extinction, see Figure 1.7 A and C), prevailing at that time. Van Valen interpreted this finding from a Darwinian perspective. If extinction is indeed an age-independent process, the demise of a taxon could therefore only result from an ‘environmental deterioration’, and a subsequent failure of the taxon to keep up within its own ‘adaptive zone’ (in ecological terms), thus subsequently losing the ‘*struggle for life*’ (Darwin 1859). The extinction rate from the survivorship curve (Figure 1.7) would therefore provide a measure of the rate of environmental deterioration. Inspired by Lewis Carroll’s *Through the Looking-Glass*, when the Queen tells Alice that “*it takes all the running [she] can do, to keep in the same place*”, Van Valen thus named his *New Evolutionary Law* the *Red Queen* hypothesis (Liow *et al.* 2011).

The implications of the Red Queen hypothesis in terms of macroevolutionary modes lie within the way this ‘deteriorating environment’ is defined. Van Valen proposed that this constant environmental deterioration would result from the fact that all species within a given assemblages are part of a ‘zero-sum game’ driven by biotic interactions, mainly competition. This means that each minor gain of fitness for a given species would negatively impact the fitness of the others, but as they evolve as well, the potential compensations of fitness loss that could have been subsequently acquired would maintain the overall fitness of the system at a constant value. Consequently, the modern appreciation of the Red Queen hypothesis is that changes in diversification rates are mainly driven by biotic factors (*e.g.*, Barnosky 2001; Benton 2009).

Several studies exemplified that biotic interactions could affect clade diversification, thus providing

support for the Red Queen as primarily conceived by Van Valen. For instance, competition most likely shaped the macroevolutionary fate of canids (Silvestro *et al.* 2015), sharks (Condamine *et al.* 2019b), antilocaprids (Nascimento & Pires 2025a), Cretaceous non-avian dinosaurs (Condamine *et al.* 2021) or gymnosperms (competing against angiosperms) (Condamine *et al.* 2020). Also, predation most likely affected the diversification of several prey and predator clades worldwide (*e.g.*, Nascimento *et al.* 2024; Nascimento & Pires 2025b; Prevosti *et al.* 2024; Solórzano *et al.* 2024; Tarquini *et al.* 2022). Another elegant example of how predation shapes clade diversification is the ‘*Escape and Radiate*’ model, proposed by Ehrlich & Raven (1964) and subsequently named by Thompson (1989). This model posits that phytophagous insects acquiring capacity to break the chemical defenses of a given plant clade undergo a rapid subsequent diversification (somewhat related to the notion of adaptive radiation, see previous sub-section) (Jousselin & Elias 2019). In response, the plant may evolve new chemical toxins, thus maintaining the selective pressure towards toxin-resistance among the insect clade (Jousselin & Elias 2019). Last, diversity-dependent diversification, namely, the set of hypotheses according to which diversification of a clade could be bounded by its own diversity (Rillo & Etienne 2022) can also provide insightful examples on how the Red Queen may shape the fate of the clades (*e.g.*, Condamine 2018; Condamine *et al.* 2019a; Etienne *et al.* 2012; Foote 2023).

Besides, although it clearly makes no doubt that the Red Queen’s predictions appropriately apply to a wide range of groups (as we intended to outline in the previous paragraph), it should be mentioned that Van Valen’s finding on age-independent extinction (Figure 1.7 B) triggered much debate in the palaeobiological community (Doran *et al.* 2006; Liow *et al.* 2011). Subsequent analyses of the fossil record of varied groups in fact often evidenced that groups could experience significant age-dependent extinction (ADE) patterns throughout their evolutionary history, preferentially towards younger species having greater extinction probabilities (*i.e.*, Figure 1.7 C) (*e.g.*, Condamine *et al.* 2019b; Finnegan *et al.* 2008; Kocáková *et al.* 2025; Saulsbury *et al.* 2024; Weppe *et al.* 2023), though noticeable studies also provided support for the opposite patterns (Figure 1.7 A) (*e.g.*, Guinot & Condamine 2023). The ADE must therefore be an additional example of the complexity of the biological reality and may selectively apply to some clades and not others. Also, as stressed by Hagen *et al.* (2018), one of the keys behind these apparent contradictory findings must lay within the taxonomic scale at which the study is conducted, most large-scale analyses of the fossil record histotically considering ranks above the species level, with ADE most likely not acting similarly at the level of species and higher taxa.

Another widely considered set of macroevolutionary hypotheses postulate that clade diversification may be impacted by phenotypical traits (Simpson 1953). The idea that phenotypical traits can influence diversification has indeed been around since the formulation of the concept of adaptive radiation, in which the brutal expansion of a clade intimately results from the emergence of a phenotypical feature (‘key innovation’, see previous sub-section) (Simões *et al.* 2016). A wealth of hypotheses have been formulated to relate clade diversification to phenotypical traits. For instance, larger body sizes have long been assumed to experience elevated extinction rates (Raup 1986). Also, pioneer work on insects showed that herbivorous lineages were more diversified than carnivorous lineages, suggesting diet-dependent diversification rates (Mitter *et al.* 1988). Subsequent methodological improvements led to the expansion of a class of diversification models known as State-dependent Speciation and Extinction (often abbreviated SSE), specifically investigating the existence of differential diversification regimes along trait categories (*reviewed in* Morlon *et al.* 2024), providing unprecedented insights into the mode of diversification for a wide range of groups. For instance, in the plant family Solanaceae, self-incompatible lineages were shown to diversify faster than self-compatible lineages (Goldberg *et al.* 2010). In Hymenoptera (wasps, ants, sawflies and bees), secondary phytophagy seems to be associated to elevated diversification rates (Blaimer *et al.* 2023). Also noteworthy, the methodological progress in

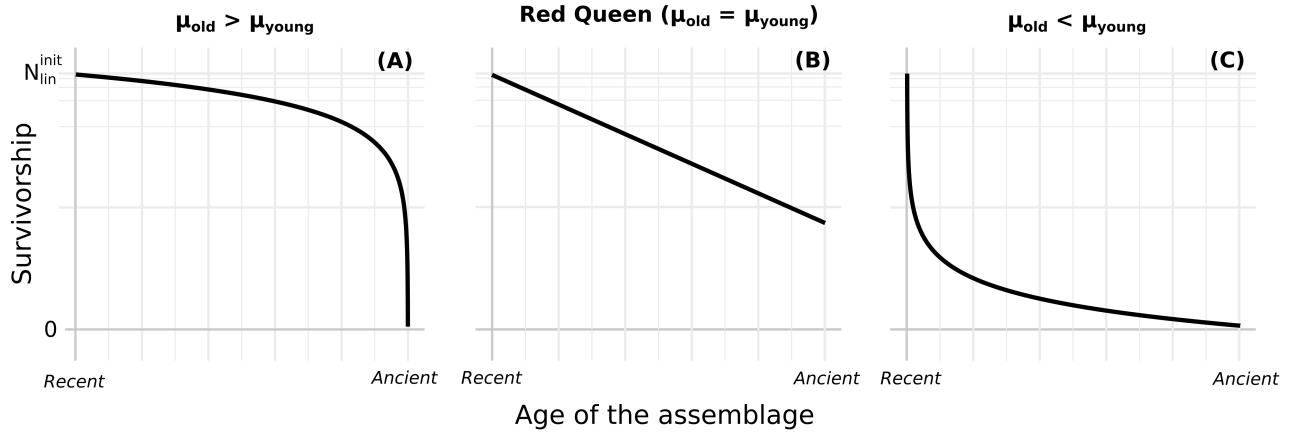


Figure 1.7: Expectations of survivorship as a function of time under three different scenarios. In all three representations, the slope of the curve represents the lineage loss per unit of time, and can thus be considered as a proxy of the extinction rate, noted μ . N_{lin}^{init} depicts the initial number of lineages. In (A), extinction rates are higher in ancient times than in recent times, suggesting that older lineages have a higher extinction rate than younger lineages (hence, $\mu_{old} > \mu_{young}$). The exact symmetrical case is shown in (C), where $\mu_{young} > \mu_{old}$. In (B), loss of lineages (hence extinction rates) does not vary across time. Extinction is thus age-independent, which is the primary evidence used by Van Valen to build the Red Queen hypothesis (Van Valen 1973). Note that the y axis has been log-scaled.

character evolution modelling (Beaulieu & O’Meara 2016) have allowed to invalidate several prevailing hypotheses regarding the effect of certain traits on diversification. Evidence from sharks (Marion *et al.* 2024) and amphibians (Liedtke *et al.* 2022) have indeed suggested that diversification was not related to reproduction mode in these clades, contrary to what was evidenced using earlier methods (*e.g.*, Mull *et al.* 2024).

Alternatively, evidence primarily from the fossil record challenged the conspicuousness of the Red Queen as the main force driving diversification dynamics. Pioneering studies indeed emphasised alternative models in which Van Valen’s Law of Constant Extinction could in fact also arise from changes in species’ physical environment (*e.g.*, Stenseth & Smith 1984; Vrba 1993). Thus, several works formulated hypotheses instead relating diversification dynamics to abiotic changes, stressing the occasional character of these physical disruptions (*e.g.*, climatic hinge, bolide impact, volcanism). As an echo to Van Valen’s metaphor, these hypotheses were summarised under the *Court Jester* model by Barnosky (2001). The Court Jester hypotheses thus “share the common tenet that changes in the physical environment, rather than biotic interactions, are the initiators of major changes in organisms and ecosystems” (Barnosky 2001). Several empirical studies provided substantial support for such a model. Diversification dynamics of a wide range of clades were indeed found to be significantly impacted by temperature changes, for instance in butterflies (*e.g.*, Condamine *et al.* 2018; Toussaint *et al.* 2025), ferns (*e.g.*, Lehtonen *et al.* 2017), marine invertebrates (*e.g.*, Guo *et al.* 2023; Malanoski *et al.* 2024), sharks (*e.g.*, Condamine *et al.* 2019b) or mammals (*e.g.*, Bekeraïté *et al.* 2025; Weppe *et al.* 2021). Mountain orogeny was also frequently illustrated among the main drivers behind clade diversification. The Himalayas indeed left significant imprints in the diversification of Apollo butterflies (Condamine *et al.* 2018) and plants (*e.g.*, Pandit *et al.* 2025). Similarly, the build-up of the Andes affected the macroevolutionary fate of mammals (*e.g.*, Pino *et al.* 2022; Solórzano *et al.* 2024;

Vallejos-Garrido *et al.* 2023), frogs and lizards (Boschman & Condamine 2022) or plants (Luebert & Weigend 2014). Additionally, major factors usually considered to impact clade diversification include palaeogeographical dynamics, with many studies supporting that view (*e.g.*, Malanoski *et al.* 2026).

As a result, it stands out that the modes of clade diversification are in fact most likely affected by a myriad of different biotic and abiotic factors (Benton 2009). The key would reside in a matter of scale (Gould 1985). Meanwhile the Red Queen is preferentially expected to occur at smaller spatio-temporal scales, the Court Jester would instead occur on longer timescales. Whether these scale differences affect the relative importance of one or the other in shaping the dynamics of clade diversification remains an open question.

1.2.2 Macro-eco-evolutionary understanding of the spatial organisation of life: insights from the Latitudinal Diversity Gradient

The uneven species richness across the globe that we outlined in Section 1.1.1, and notably, the striking pattern of increasing species richness with decreasing latitude famously known as the ‘Latitudinal Diversity Gradient’ (LDG), can be understood under a macroevolutionary perspective involving ecological processes. Several hypotheses (*reviewed in* Pontarp *et al.* 2019) have been proposed to explain such a pattern, some of which imply the interplay between diversification rates and tropical environmental conditions (Tietje *et al.* 2022).

In the footsteps of Stebbins (1974), the ‘cradle’ hypothesis posits that tropical lineages would have larger speciation rates than extra-tropical lineages. Higher speciation rates for tropical lineages may be predicted from tropic’s greater environmental heterogeneity (Stein *et al.* 2014), increasing available niche space. Conversely, one can expect speciation rates to be lower outside the tropics, where climate seasonality is more pronounced. In addition, the ‘Metabolic Theory of Ecology’ (Brown *et al.* 2004) can also predict that tropical lineages would have larger speciation rates. According to the latter, the higher energy availability for the tropics (higher temperature) would locally increase tropical species’ metabolic rate (Allen *et al.* 2002), yielding larger mutation, and thus, speciation rates (Allen *et al.* 2006; Lanfear *et al.* 2010). Also, as summarised by Mittelbach *et al.* (2007), tropical regions’ greater species richness has been suggested to enhance the importance of biotic interactions (Schemske *et al.* 2009). Consequently, since the ‘Red Queen’ hypothesis (Section 1.2.1) postulates that biotic interactions would enhance speciation, this would establish a positive feedback loop within the tropics, where increasing diversity would yield increasing speciation rates. This ‘cradle’ hypothesis is often contrasted with the ‘museum’ hypothesis, instead placing the emphasis on the fact that tropical lineages would have lower extinction rates. The ‘museum’ hypothesis may be a consequence of the ‘tropical stability’ hypothesis, assuming that the tropics have remained climatically stable across geological times, and that this stability would have made tropical lineages less prone to extinction (Wallace 1878). In addition, it might be expected that the aforementioned environmental heterogeneity offered by the tropics could buffer tropical lineages against eventual environmental changes.

A consensus emerged from late 2000s–2010s studies testing the extent to which diversification rates could explain the observed LDG, where the tropics may actually both act as a cradle and a museum of diversity – some works also emphasised the role of tropical-to-extratropical dispersal as a side cause behind the Great American Biotic Interchange, see Section 1.4.2. This view has been contrastingly supported in the recent literature. Nevertheless, joint analyses of fossil and phylogenetic data for several tetrapod clades, using cutting-edge methodological frameworks, provided support for elevated speciation rate outside the tropics as a likely mechanism for the LDG, thus instead favouring the ‘museum’ hypothesis (Meseguer & Condamine 2020; Quintero *et al.* 2023). Conversely, a recent

study focusing on a massive sample of the seed plant diversity emphasised that no causal relationship could be established between diversification rates and species richness (Tietje *et al.* 2022). Given these elements, it seems like the study model, the primary source of data (*e.g.*, fossils vs. dated phylogenies) and the methodology used to estimate diversification rates (see Section 1.3.1) affects upstream interpretations of the interplay between diversification rates and spatial patterns of species richness.

However, the notions of ‘cradle’ and ‘museum’ of biodiversity have recently been challenged by Vasconcelos *et al.* (2022), emphasising a number of limitations for the way they are often considered in the literature. Among them, as several factors driving speciation, such as climatic stability/heterogeneity, may also drive extinction, they argue that it would sometimes make more sense to consider turnover rates (see Glossary) rather than speciation and extinction rates individually. Also, they point out that extant centres of biodiversity with a great concentration of ‘basal’ lineages are not necessarily centres of origin, thus challenging both the way previous works actually interpreted the notion of museum and cradle of diversity. The latter nuance echoes previous works of the same authors emphasising the need to include hidden states in models of character evolution, marking a turning point in macroevolution (*e.g.*, Beaulieu & O’Meara 2016).

Alternative explanations of the LDG not involving diversification rates have also been proposed. These either refer to species’ ecological limits (*e.g.*, carrying capacity, size of the available area, palaeogeography) or time left for species to diversify (assuming tropical stability would have left more time for species to diversify within than outside the tropics, and attested by fossil and phylogenetic evidence of lineage age distribution in and outside the tropics) (Pontarp *et al.* 2019). As stressed by Mittelbach *et al.* (2007), these three classes of hypotheses (diversification-, ecological- or time-dependent) are not necessarily mutually exclusive. Conversely, a fourth set of hypotheses emphasised the likely implication of pure contingency in shaping the LDG, notably through the use of null models (Arita & Vázquez-Domínguez 2008). Recently, insights from eco-evolutionary simulations (see Section 1.3.2) evidenced that null models – in which the modelled patterns do not rely on particular eco-evolutionary mechanisms (Gotelli & McGill 2006) – could indeed match empirical LDG patterns, although generally worse at explaining empirical distributions data than model incorporating mechanistic constraints on species, in particular, climate and palaeogeography (Hagen *et al.* 2021; Lorcery *et al.* 2025).

1.3 Analytical tools to study large-scale biodiversity patterns

This section is aimed to provide a broad outline of some methodological means to investigate the patterns and questions raised in the previous sections. Please note that each chapter of this thesis has its own *Methods* section.

1.3.1 Macroevoolutionary approaches to study how clades thrive and go extinct

The birth–death process

In macroevolution, the dynamics of clade diversification are usually modelled by a birth–death process (of course, this only holds for *model-based methods*, not for *count methods*, see next section) (see Tavaré 2018 for an exhaustive outlook). This stochastic process, borrowed from the field of population dynamics, is a classical example of continuous-time Markov process. Consequently, under this model, the number of species within a clade at any step i (here noted N_i) only depends on the number of

species in the previous step (N_{i-1}). In addition, N_i is ruled by two processes independently occurring at the scale of every individual lineages. Each lineage can indeed give birth to a new lineage (*speciation* event), or die (*extinction* event). In a birth–death process, speciation and extinction are modelled as Poisson point processes, and their respective intensities (also called rates) are usually noted λ and μ , expressed in number of events per lineage per unit of time. Thus, the birth–death process makes the strong assumption that speciation and extinction are instantaneous events, which may not be appropriate, particularly in the case of speciation (see Morlon *et al.* 2024 for further discussion).

Kendall (1948) generalised the constant-rate birth–death process by allowing λ and μ to be functions of time. Half a century later, this foundational mathematical development spurred significant theoretical and computational innovations in the field of macroevolution (again, see Morlon *et al.* 2024 of an extensive review). Notable improvements of the birth–death process for the inference of clade diversification have involved the incorporation of fossil and extant sampling parameters (e.g., Heath *et al.* 2014; Stadler 2010), a fossilisation parameter (e.g., Silvestro *et al.* 2014b; Stadler *et al.* 2018); have allowed for more precise estimations of the timing of diversification rate variations (e.g., Kopperud & Höhna 2025; Quintero *et al.* 2024; Rabosky 2014; Silvestro *et al.* 2019); state-dependent diversification rate variations (e.g., Beaulieu & O’Meara 2016; Maddison *et al.* 2007); rate variations between clades (Morlon *et al.* 2011) or even lineages (Maliot *et al.* 2019; Quintero *et al.* 2024); rate variations along external abiotic and/or biotic correlates (e.g., Etienne *et al.* 2012; Lehtonen *et al.* 2017; Licht *et al.* 2026; Silvestro *et al.* 2015) or combinations of several of these features (e.g., Hauffe *et al.* 2024; Quintero *et al.* 2023).

Quick overview of the philosophy behind the Bayesian approach to parameter inference

Today, some diversification models – and notably those applied over this thesis – are implemented in a Bayesian framework. It is thus appropriate to clarify some key notions of Bayesian inference before moving on to the models *per se*.

Let’s say one wants to explain some data $\mathcal{D}$ using a statistical model $\mathcal{M}$, going along a set parameters θ . The type of data is not important here, but to make it more anchored in the overall framing of this doctoral work, it can be considered that $\mathcal{D}$ is a set of fossil occurrence data (defined by their age), and that $\mathcal{M}$ is a diversification model, thus including two key parameters (origination and extinction rates, respectively λ and μ), so that $\theta = \{\lambda, \mu\}$ (note that θ may include any additional parameter related any other process incorporated within $\mathcal{M}$, such as sampling, see previous section).

The goal of any model-based statistical inference is to explain $\mathcal{D}$ with $\mathcal{M}$. This aim may be achieved with two kind of approaches implying fundamentally distinct ways of reasoning. The first one consists of finding the set of parameter values θ^* maximising the probability of the data given the model, also called *likelihood*, noted $\mathcal{P}(\mathcal{D}|\theta)$. This is typically done in *Maximum-Likelihood* inference, and results in a single estimate of parameter values (θ^*), also called point estimate. Conversely, the initial goal may also be fulfilled by focusing on the probability of the model given the data, called *posterior probability* and noted $\mathcal{P}(\theta|\mathcal{D})$, which is the core of *Bayesian* inference. The Bayes’ theorem links the likelihood and the posterior probability in the following way: $\mathcal{P}(\theta|\mathcal{D}) = \frac{\mathcal{P}(\mathcal{D}|\theta)\mathcal{P}(\theta)}{\mathcal{P}(\mathcal{D})}$.

$\mathcal{P}(\theta)$ is called *prior probability* (often also simply referred to as ‘prior’). It reflects potential *a priori* knowledge on model parameters, which constitutes one of the strengths of the Bayesian approach. In Bayesian inference, it is indeed possible for the user to inform the model, and thus, to sort of guide the parameter inference step. That being said, priors have also been the subject of many of the critics formulated against Bayesian inference, since they may also be viewed as a way to bias the inference

step. The case of informative *vs.* non-informative priors has also been much debated (Lemoine 2019), but will not be detailed here.

Also, $\mathcal{P}(\mathcal{D})$ represents the *marginal probability* of the data. It could be obtained by integrating over all possible parameter combinations ($\mathcal{P}(\mathcal{D}) = \int_{\theta} \mathcal{P}(\mathcal{D}|\theta)d\theta$), but this is often computationally hard to achieve in the real world. To circumvent this apparent hindrance to posterior probability calculation, Markov Chain Monte Carlo (MCMC) algorithms are used during Bayesian parameter inference, relying on computing posterior ratios between two sampling steps within the parameter space, thus avoiding to calculate marginal probabilities (see Barido-Sottani *et al.* 2024 for clear in-depth explanations).

Overall, one of the main forces of Bayesian inference is that it estimates parameter distributions, thus offering a direct quantification of the uncertainty around parameter estimation, strikingly contrasting with Maximum-Likelihood’s point estimates – requiring bootstrapping to get an estimate of uncertainty.

Fossil occurrence-based methods to infer the dynamics of clade diversification

The first estimations of origination and extinction rates in macroevolution have relied on the only direct source of evidence of life in the past, that is, the fossil record (*e.g.*, Newell 1952). These may be grouped into two different classes.

Count methods have been the first historically applied. As their name indicates, these have originally consisted of approximating origination and extinction rates within a given time bin with raw counts of the number of taxa appearing or disappearing within such a bin (*e.g.*, Newell 1952; Webb 1976). However, the palaeobiological community’s growing concern about the caveats of the fossil record (Section 1.1.4) challenged the accuracy of such estimates. Notably, the characterisation of the ‘pull of the recent’ and the ‘Signor-Lipps’ effects (Section 1.1.4) implied to re-think the structure of these inherently biased sampling schemes, triggering substantial methodological improvements within that set of methods. Several of them have consisted of (1) relying on consecutive time windows sliding throughout the lifespan of a given taxon and (2) applying some criteria indicating whether or not such a taxon should be incorporated within a given bin with respect to the other windows. Bin-specific origination and extinction rates were thus computed based on equations developed out of these adapted counting schemes (reviewed in Alroy 2014). The most popular of these methods include the ‘Boundary Crosser’ (Foote 2000), the ‘two-timer’ and ‘three-timer’ (Alroy 2008) or the ‘Gap-Filler’ (Alroy 2014) equations.

On the other hand, over the past ten years, *model-based methods* relying on a birth–death process (see previous sub-section) have been an increasingly more frequent means to estimate diversification rates from fossil occurrences. To account for some of the key biases of the fossil record, this set of methods may explicitly incorporate those biases as probabilistic processes. For instance, PyRate (Silvestro *et al.* 2014a), probably the most popular of these frameworks (and intensively used over this thesis, see Chapter 1 and 2) jointly infers diversity, origination and extinction rates along with a preservation process, explicitly modelling fossilisation bias (Silvestro *et al.* 2014b). In addition, PyRate offers the possibility to account for (1) potential lineage-heterogeneous preservation intensities (Section 1.1.4) by means of a gamma model – just like for substitution rates in phylogenetic inference – and (2) the time-heterogeneity of the preservation process by allowing preservation rates to vary in a set of user-defined time bins (Section 1.1.4) (Silvestro *et al.* 2019). In addition, PyRate has been implemented in a Bayesian framework, thus offering a direct access to parameter uncertainty (see previous section).

PyRate has known substantial improvements throughout the past 10 years, that allowed to test for

increasingly complex macroevolutionary hypotheses. Notable progress have for instance involved the possibility to assess for the effect of environmental variables on rate variations (Lehtonen *et al.* 2017; Silvestro *et al.* 2015) or to test for age-dependent extinction (see Section 1.2.1) (Hagen *et al.* 2018). More recently, using deep-learning, the Birth–Death Neural Network (BDNN) model has extended PyRate’s framework by allowing for lineage-time-specific diversification rates estimated as non-linear combinations of a user-defined set of environmental predictors, thus significantly expanding the set of testable macroevolutionary scenarios (Hauffe *et al.* 2024).

However, PyRate’s main shortfall is that it makes the assumption of complete lineage sampling and therefore does not incorporate sampling bias in its framework. This is problematic since empirical datasets, and particularly fossil occurrences, are subject to high sampling heterogeneities (*cf.* Section 1.1.4). Methodological efforts have been invested to account for this limitation. Some include applying spatial standardisation procedures to the input dataset and/or incorporate a Bernoulli process at the base of the diversity inference to account for unsampled lineages (Flannery-Sutherland *et al.* 2022). More recently, training a deep-learning algorithm with a large number of simulated fossil datasets has provided promising results for leveraging the impact of sampling bias on taxon richness across time (Cooper *et al.* 2024). Also, other occurrence-based diversification models exist, leveraging the aforementioned incomplete sampling shortfall, such as the treeless Fossilised Birth–Death range model (Stadler *et al.* 2018; Warnock *et al.* 2020), explicitly incorporating a sampling-through-time parameter (Stadler 2010).

Phylogeny-based methods to infer the dynamics of clade diversification

As this thesis will not apply such methods, only a brief outline will be given here.

Phylogenies are commonly-used representations of the relationships between a given set of taxa by means of acyclic undirected graphs (‘phylogenetic trees’), in which terminal nodes (‘leaves’, at the end of the terminal edges, ‘branches’) represent the taxa, and the initial node (‘root’) represents the common ancestor to each of these taxa. Every internal node has the particularity of being dichotomous, and thus, is connected by exactly three branches. Dated phylogenies, in which branch lengths are proportional to divergence times between species, have long been recognised to contain precious information about the dynamics of clade diversification (Morlon 2014). With a simple lineage-based model in which, at every time step, lineages could speciate or go extinct with given probabilities, Raup *et al.* (1973) simulated phylogenies with impressively similar features to empirical trees. In fact, without explicitly naming it, they demonstrated that it was possible to model the process of diversification with a birth–death model (see previous sub-sections), which was further formalised two decades later (Nee *et al.* 1994). From these outstanding contributions, a wide array of increasingly-complex methods have been developed, reviewed in Morlon *et al.* (2024).

A subtle point of the approach lies in that, in macroevolution, the birth–death process is the preferred model to build trees (other tree-building methods exist, such as the coalescent, but their use is not common in macroevolution, see Cornuault *et al.* (2025)). On the other hand, birth–death models may also be fitted to already-reconstructed dated trees to make some inferences on the most likely diversification processes explaining the tree, for instance, using model selection (*e.g.*, Condamine *et al.* 2019a; Lewitus *et al.* 2018). Conceptual issues may arise from this subtlety, for instance, when fitting birth–death models with time-variable rates to trees reconstructed under a constant-rate birth–death process.

It should also be mentioned that estimating diversification rates from phylogenies has been inten-

sively debated in the community. Particularly, several studies raised the issue that the shape of extant phylogenies could be consistent with an infinity of speciation and extinction rate combinations, and thus, that these two parameters were not identifiable (the problem coming from extinction rate estimation using extant-only phylogenies, greatly subsampling the evolutionary history of the focal clade) (Louca & Pennell 2020; Rabosky 2010). One solution to circumvent such a critical bias lies in the combination of extant and fossil data, for instance, in an approach known as the Total-Evidence Dating (Gavryushkina *et al.* 2017; Pyron 2011; Warnock & Wright 2020). This powerful framework relying on a tree-building process known as the Fossilised Birth–Death (FBD) process (Mulvey *et al.* 2025) was indeed shown to leverage the aforementioned identifiability issue (Truman *et al.* 2024). However, such models are computationally intensive. In addition, the two sources of information used by the FBD applied to Total-Evidence Dating (molecules and morphology, but not that the FBD process may be used to carry out phylogenetic reconstructions from molecular-only datasets) are sometimes difficult to reconcile as not necessarily subject to the same evolutionary constraints. Also, these two sources of data are not always accessible, *e.g.*, in groups with no known fossil record. Consequently, although it stands out as an ideal avenue to infer diversification from phylogenies, such a framework may be hard to apply in practice.

1.3.2 Integrating ecological processes with evolutionary forces: mechanistic eco-evolutionary simulations

The traditional approach to eco-evolutionary modelling has consisted of fitting models to data (as seen for diversification models, see Section 1.3.1). Such an approach is of course inherently affected by the existing biases within the data, intensively discussed in the previous sections (Section 1.1.4), and has left little room for casual testing of the actual mechanisms underlying the observed patterns. The use of simulation models has opened worthy avenues to overcome these issues (Gotelli *et al.* 2009). Among them, spatially-explicit population-based mechanistic eco-evolutionary models (MEEMs) have proven highly relevant to tackle issues related to the emergence and the dynamics of biodiversity patterns across deep time (Hagen 2023).

MEEMs combine a gridded landscape made up with a set of user-selected variables (*e.g.*, palaeoclimate) with a set of eco-evolutionary rules to simulate virtual species. The use of simulated data allows MEEMs to leverage any caveat coming along with empirical data. Since the very first introduction of simulation-based models in ecology and evolution by Gotelli *et al.* (2009), several MEEMs have been released, offering unprecedented insights into the mechanisms at play to fashion the spatio-temporal dynamics of biodiversity (*e.g.*, Leprieur *et al.* 2016; Rangel *et al.* 2018; Saupe *et al.* 2020; Skeels *et al.* 2023). One of the most popular is the general engine for eco-evolutionary simulations, also called *gen3sis*, that has been used over this thesis (Chapter 3)(Hagen *et al.* 2021).

In *gen3sis*, virtual species are modelled with individual populations (one population per grid) evolving independently in the pre-defined landscape, following a set of four rules. (1. *Dispersal*) Populations may disperse from one cell to another according to a user-defined dispersal kernel (often a Weibull distribution). (2. *Speciation*) If two populations happen to be disconnected, as long as they remain so, they will accumulate divergence at a user-defined rate across simulations steps. If inter-population divergence falls above a user-defined speciation threshold, these two populations will convert into two distinct species. Consequently, the only speciation mode that *gen3sis* is so far able to simulate is allopatry (see Glossary). (3. *Environmental filtering*) Species may be given environmental tolerances (or niches) along the variables included in the landscape, determining their level of fit to local environment at the cell scale. (4. *Trait Evolution*) Species-specific niche are defined following key traits

(*e.g.*, temperature optimum, breadth, competitive ability) that may themselves evolve across simulation steps, or be fixed. Trait evolution may be driven by environmental changes, thus offering the possibility to test for different ecological scenarios (null or neutral vs. adaptive).

Consequently, by directly specifying the eco-evolutionary processes mediating species' ability to maintain within a landscape, *gen3sis* (as well as any other MEEM) stands out as a powerful and flexible tool to mechanistically assess a wide range of hypotheses on how biodiversity patterns emerge and are maintained over time.

1.4 A strange biodiversity on a strange land : the case of mammals in South America

“[The Neotropical] region includes South America, the Antilles and tropical North America; and it possesses more peculiar families of vertebrates and genera of birds and mammalia than any other region” (Wallace 1876)

1.4.1 South America across the Cenozoic: elements of palaeogeographical and palaeoenvironmental context

Due to its unique physical geography, South America offers an immense diversity of habitats, from tropical alluvial plains, dry highlands, grasslands to the top of the second highest mountains on Earth (Figure 1.8). Understanding the build-up of South American landscapes is vital to disentangle the build-up of its astonishing biodiversity.

South America had a very singular palaeogeographic history. Akin to Australia in the present-day, the continent spent most of the Cenozoic completely isolated from other landmasses (Scotese & Wright 2018). In terms of faunal evolution, such an isolation had strong implications, that will be further detailed in Section 1.4.2.

Probably the main feature that stands out as one stares at the present-day map of South America is its marked topographical heterogeneity. The eastern side of the continent indeed appears topographically homogeneous, despite the existence of a few rolling areas like the Brazilian Highlands or the Guianian Shield (both composed of billion-years-old cratons, Figure 1.8). On the contrary, with the Andes stretching North-to-South over 8000 km, the western margin of South America is considerably more topographically heterogeneous. The Andes, including more than 70 peaks above 5000 m and two out of the three highest plateaus on Earth, are the result of the subduction of the Nazca plate beneath the South American plate (Ramos 2009). Although the Andean subduction system started way earlier, the Andean orogenetic (*i.e.*, mountain-building) cycle began in the late Early – early Late Cretaceous (Horton 2018; Ramos 2009). This uplift intensified from the Late Miocene onwards, showing a pronounced northwards-propagating diachrony (Boschman 2021; Gianni *et al.* 2020) (Figure 1.10). The eastern part of the Andes (matching the Andean foreland basins, Figure 1.8), offering a wide altitudinal range and a tropical climate, is arguably the most species-rich region in the world (Figure 1.2), and the origin of its huge local diversity has puzzled biogeographers since the times of Humboldt (Antonelli *et al.* 2018; Rahbek *et al.* 2019a, b). In addition, it should be mentioned that a significant amount of low-latitude fossil evidence unearthed in the recent years came from this foreland basin (Antoine *et al.* 2017) (Figure 1.6, 1.8 and 1.9).

The palaeogeographical, palaeoclimatic and palaeoenvironmental history of South America during the Cenozoic was tightly related to the pace of the Andean uplift and its spatial dynamics (Hoorn *et al.* 2010). A wealth of evidence indeed attests that the elevation of this mountain chain profoundly re-organised continental climate, notably since it represents a topographic barrier to atmospheric circulation, and thus, aridified some places and intensified precipitations in others (Blisniuk *et al.* 2005; Hartley 2003). These effects, of increasing intensity as the Andes got higher (notably since the Middle Miocene, Figure 1.10), combined with global-scale temperature dynamics greatly shaped the landscapes at the continent scale, and notably the rise of open landscapes at the expense of closed forests (*e.g.*, Palazzesi & Barreda 2012).

The Andean uplift also exerted a major control on continent-scale hydrographic patterns, in turn strongly impacting the structure and composition of faunal and floral assemblages locally. From the Early to the late Middle Miocene (*ca.* 20–10 Ma), a vast wetland known as the ‘Pebas MegaWetland System’ (PMWS, Figure 1.10) covered Western Amazonia, occupying significant portions of what is now Colombia, Ecuador, Peru, Venezuela and Brazil (Hoorn *et al.* 2022a). There is clear evidence supporting that, together with eustatism, the Andean uplift was the primary control of the emergence and sustaining of this system, mostly by causing subsidence and locally increasing precipitation rates (Hoorn *et al.* 2022a and literature therein). The PMWS was characterised by a complex mixture of interconnected aquatic environments, such as lakes or swamps (somewhat analogous to what may be found in the Pantanal region in Brazil, see Marivaux *et al.* (2020a) and Hoorn *et al.* (2022b)), and was home to a unique biodiversity, very distinct from what is recovered from extant assemblages in the region (*e.g.*, Antoine *et al.* 2017; Salas-Gismondi *et al.* 2015; Tejada-Lara *et al.* 2015). The demise of the PMWS was most likely also a by-product of the Andean uplift. Similarly, there is strong evidence supporting that the Late Miocene marine introgression that flooded parts of what is now Northern Argentina, Brazil and Southern Bolivia, known as the ‘Paranense sea’ (named after the region of Paraná, see Figure 1.8 and 1.10) was primarily controlled by the Andean uplift (Hernández *et al.* 2005). The ecosystems generated by the Late Miocene demise of the Paranense sea are suspected to have been favoured lineage diversification, as shown for caviomorph rodents (*e.g.*, Candela *et al.* 2012).

Last, South American ecosystems were also immensely impacted by the closure of the isthmus of Panama, regarded as an incontestable facilitator for the inter-American faunal admixture known as the ‘Great American Biotic Interchange’ (GABI, see Figure 1.10 and Section 1.4.2) (Jaramillo 2018). The timing of the completion of an overland connection between the Americas, having profound implications for the temporal calibration of the GABI, has been widely debated. The age of 3.5 Ma has long been used as a benchmark (Bacon *et al.* 2015) but geological (Coates & Stallard 2013; Montes *et al.* 2015) and biological (Bacon *et al.* 2015) evidence also suggested an earlier age, raising some criticism (Lessios 2015; Marko *et al.* 2015; O’Dea *et al.* 2016). Nowadays, integrating data from several areas of Earth and Life sciences resulted in a consensus in which deep-water flow in the isthmian region ceased with the closure of the Central American Seaway by the late Middle Miocene (*ca.* 10 Ma). After this date, it is generally accepted that the Pacific and the Caribbean still exchanged shallow water until the Late Pliocene (*ca.* 3.5–2.8 Ma), marking the emergence of a land bridge – the so-called isthmus of Panama – that most likely persisted until nowadays (Jaramillo 2018; O’Dea *et al.* 2016).

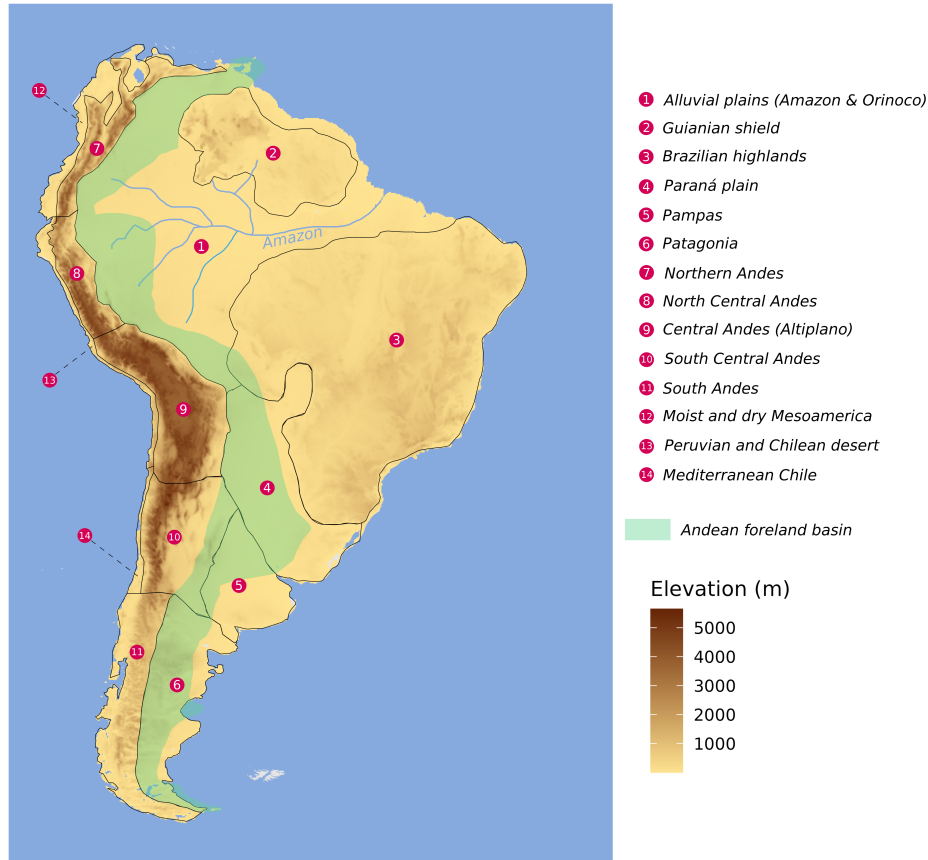


Figure 1.8: Extant physiography of the South American continent. 14 among the main (eco-) physical regions of the continents were represented on top of the present-day topography of the continent, from Boschman (2021). The approximate contour of the Andean foreland basin has been highlighted in light green.

1.4.2 'Splendid isolation': the captivating history of South American mammals across the Cenozoic

History of mammal palaeontology in South America: a brief overview

Some of the historical elements developed here may be found in further detail in Manias (2023).

South American mammal palaeontology sadly started as a colonisation by-product, as European explorators plundered Patagonian soils, intrigued by skeletal remains trapped in the rocks. In the late 18th century, some of the fossils brought back to Europe provoked curiosity among the leading figures of the emerging field of comparative anatomy. As a result, in 1796, Georges Cuvier described the famous *Megatherium americanum*, and proposed this six-meter-beast to be a sloth. A few decades later, in 1836, Charles Darwin, back from his five-years journey on board the HMS Beagle, brought back to England a wealth of fossil material (in particular from the region of Punta Alta, Patagonia – Figure 1.9). These fossil remains were given to the renowned palaeontologist Richard Owen, who described, in 1837, many emblematic fossil mammals from South America, including the car-sized

armadillo *Glyptodon*, the giant sloth *Mylodon*, the giant camel-like native ungulate *Macrauchenia* or the rhino-sized native ungulate *Toxodon*.

The second half of the 19th century marked the beginning of South American mammal palaeontology made by scientists from South America, or more precisely, from the Republic of Argentina. At that time, natural sciences, as well as any other field contributing to sharpening knowledge on the country's own territory, were envisaged as nation-building, and the Southern part of the country (Pampas and Patagonia provinces) had long been recognised to overflow with fossil remains of extinct animals. Consequently, important efforts were devoted to collect and shed light on the country's archaeological and palaeontological patrimony, centralised within national museums, such as the “Museo Nacional de La Plata”, headed by the prominent explorer and natural scientist Franciso Moreno in the 1870s.

South American mammal palaeontology owes a lot to the labour of the patriotic Argentinian palaeontologist Florentino Ameghino. Relying on fossil material mostly collected during field expeditions led by his brother Carlos, Florentino Ameghino produced a remarkable quantity of systematic palaeontological work, and his written productions are regarded as essential building-blocks for South American palaeomammalogy. These are probably best exemplified by his foundational monographies, such as “*Contribución al conocimiento de los mamíferos fósiles de la República Argentina*” (Ameghino 1889), in which several hundreds of taxa from the Argentinian fossil record are described. Argentina's fossil richness, together with the aforementioned efforts to enlighten its natural patrimony, made that South American palaeomammalogy long remained essentially constrained to the country, hence the striking spatial clustering of the fossil occurrences in the Southern cone depicted on Figure 1.9. As recalled by Simpson (1940), in the first half of the 20th century, knowledge of Cenozoic mammal remains outside Argentina was very scarce. Many of the most significant discoveries of non-Argentinian mammal-bearing fossil localities (as those highlighted on Figure 1.9) have indeed occurred throughout the last 80 years, with notable exceptions of long-recognised rich fossil-bearing localities, such as Tarija, in Southern Bolivia (Boule 1920).

In the 1930s, the young U.S. vertebrate palaeontologist George G. Simpson, later to become the renowned architect of the Modern Synthesis of Evolution (Futuyma 2015), carried out his first expeditions in South America, intrigued by the continent's peculiar fossils. Simpson further developed a strong interest towards understanding the evolution of South American mammals, that animated him all over his career (e.g., Simpson 1940, 1950, 1967; 1970). In 1980, he produced an outstanding synthesis of South American mammalian evolution across the Cenozoic, that is still used as a reference today (Simpson 1980). In his book, he proposed that the tempo of South American mammalian evolution could be summarised by three temporally-distinct ‘faunal strata’, that will be briefly outlined in the subsequent sections (Figure 1.10).

First stratum: endemic radiations (Late Cretaceous–middle Eocene)

Simpson's first stratum characterises the early phase of South American mammal evolution in the aftermath of the continent's isolation (Figure 1.10). It is thought to have lasted from the Late Cretaceous/early Palaeocene to the late middle Eocene. Mammal at that time are often referred to as the continent's ‘original inhabitants’. These would include endemic metatherians (Figure 1.11 A), native ungulates (Figure 1.11 C), xenarthrans (Figure 1.11 B), as well as a few remnant non-therian lineages (e.g., Gondwanatheria).

The oldest Cenozoic South American land mammalian assemblages date back to the early Palaeocene. These have been recovered from the Punta Peligro (Argentina) and Tiupampa (Bolivia)

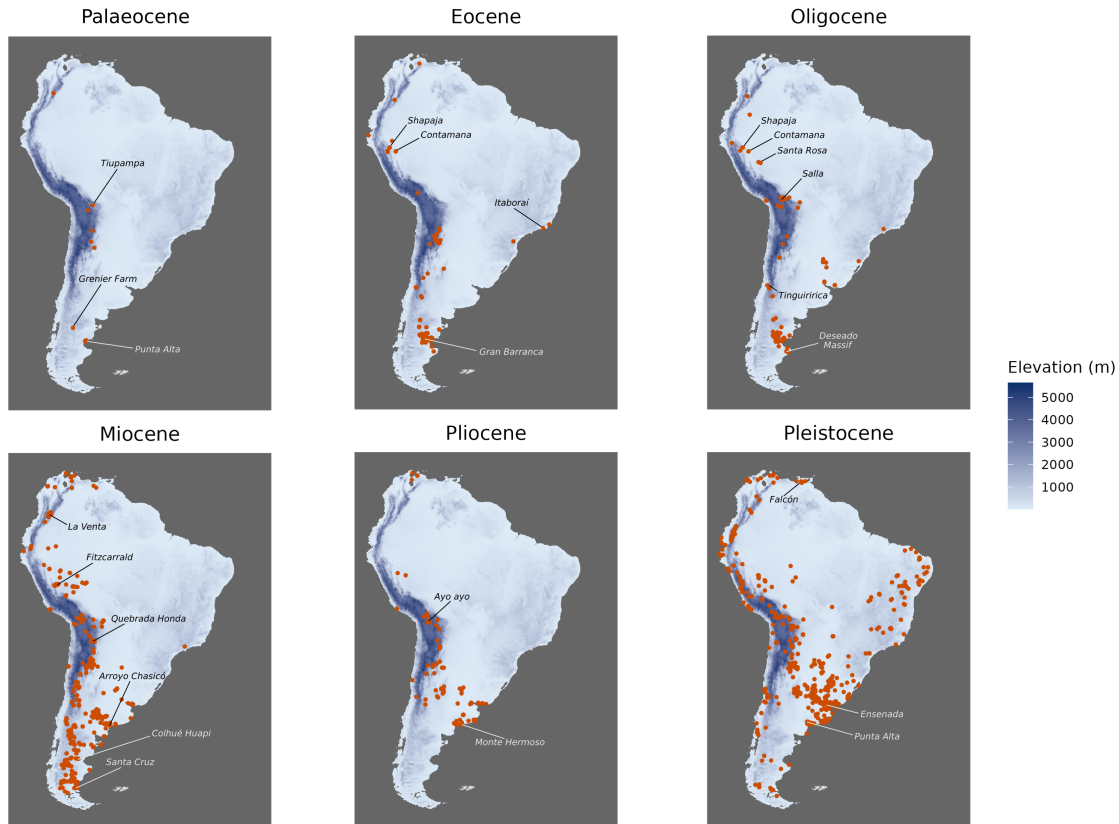


Figure 1.9: Distribution of the South American terrestrial mammal fossil record across geological epochs (Holocene excluded), on top of the continent's present-day topography ([Boschman 2021](#)). Several keystone mammal-bearing localities have been highlighted.

localities (Figure 1.9), each leading to the recognition of the oldest intervals of the South American Land Mammal Age (SALMA) scale, namely, the Peligran and the Tiupampan (Figure 1.10). Biochronological evidence suggests the Tiupampan SALMA to be older than the Peligran SALMA, and corroborate an age interval of 64.5–64 Ma for the Tiupampan fauna ([Gelfo *et al.* 2009](#)). An additional occurrence of pre-Tiupampan therian mammal was recovered from the Grenier Farm locality (Figure 1.9), and was assigned to the extinct metatherian order Polydolopimorphia ([Goin *et al.* 2006](#)).

The faunal composition of the Tiupampan and Peligran assemblages delivered precious insights into the early Cenozoic history of South American mammals. The great metatherian diversity (Figure 1.11 A), together with the presence of representatives from several ungulate-related lineages typically found in the Northern hemisphere (formerly grouped into the ‘condylarths’), suggested that South American mammalian assemblages received significant faunal supply from North America between the late Cretaceous and the early Palaeocene ([Pascual 2006](#)). Geological evidence corroborated the possibility for intermediate episodic connections between the two continents in that interval ([Pascual 2006](#)), possibly also involving the Caribbean islands ([Viñola-López *et al.* 2022](#)). As an echo to the name ‘Great American Biotic Interchange’ (see Section 1.4.2 and Chapter 3), this keystone biotic exchange for the architecture of Simpson’s first stratum was thus termed ‘First American Biotic Interchange’ (FABI) ([Goin *et al.* 2012a](#)).

Along with these therian lineages of North American affinity, non-therian lineages persisted in South America at least until the mid-Palaeogene, thus well after the Cretaceous–Palaeogene boundary (Goin *et al.* 2012b). The most fascinating example of non-therian lineage persistence known to date is probably the enigmatic *Necrolestes*, from the Santa Cruz Formation (early Miocene, Santacrucian SALMA, see Figure 1.10), identified as a remnant of the clade Meridiolestida, a mammal group that thrived from the Late Jurassic up until the Late Cretaceous (Chimento *et al.* 2012; Rougier *et al.* 2012).

Younger assemblages characteristic of Simpson’s first stratum notably include the São José de Itaboraí (often abbreviated Itaboraí) fauna (Figure 1.9), one of the most diverse in the Palaeogene of South America. It led to the Itaboraian SALMA, approximately dating back to 53 Ma (Figure 1.10). The Itaboraian fauna is still dominated by metatherians (Goin *et al.* 2016). It also revealed that all the higher South American Native Ungulate (SANU) lineages were clearly differentiated at that time (Figure 1.11 C). More importantly, the Itaboraian fauna yielded the oldest known occurrence of armadillo (Xenarthra: Cingulata) (Bergqvist *et al.* 2004), thus documenting the appearance of Xenarthra in the fossil record (Figure 1.11 B). The late appearance of xenarthrans in the fossil record remains an enigma for South American mammal palaeontology, and has been referred to as the ‘xenarthran mystery’ (Goin *et al.* 2016). Indeed, as suggested by their name meaning “weird articulations”, xenarthrans, today represented by sloths, anteaters and armadillos, share very derived morphological characters that are not found in any other known eutherian mammal lineage. Molecular evidence suggests that they form a clade with the African group Afrotheria, called ‘Atlantogenata’, literally meaning “generated by the Atlantic ocean”, and thus, would have diverged from their sister-group following the break-up of Gondwana, some 120 Ma (Waddell *et al.* 1999). Why armadillos only appear in the fossil record *ca.* 53 Ma, and sloths in the Late Oligocene (see next section), whereas their divergence with other South American eutherians largely predates the earliest known mammal assemblages found on the continent thus remains an open question.

Younger assemblages of the first stratum progressively record a turnover in the relative proportion of metatherian vs. eutherian mammals in the communities, with an increasing dominance of eutherians, that several authors related to the gradual Eocene climate cooling from *ca.* 50 Ma onwards (Croft 2016; Goin *et al.* 2016; Westerhold *et al.* 2020). Later in the Eocene, land mammal assemblages will know go through a series of gradual changes that will progressively mark the transition with the subsequent faunal stratum, detailed thereafter.

Second stratum: Trans-atlantic migrations and faunal ‘modernisation’ (late Eocene–Late Miocene)

Most specialists agree that major re-organisations occurred within South American mammal assemblages near the Eocene–Oligocene boundary (that will be more thoroughly covered in Chapter 1), marking the onset of Simpson’s second faunal stratum (Figure 1.10). The distinction between the second and the first strata lies in a three-part faunal ‘modernisation’ (Flynn & Wyss 1998): (1) emergence of morphological innovations among “original inhabitants”, (2) extinction of basal, ‘archaic’ lineages and (3) higher-level taxonomic origination within “original inhabitants”, alongside the arrival and radiation of hystricognathous rodents (Rodentia: Hystricognathi: Caviomorpha) and anthropoid primates (Primates: Anthropoidea) in South America (Figure 1.10).

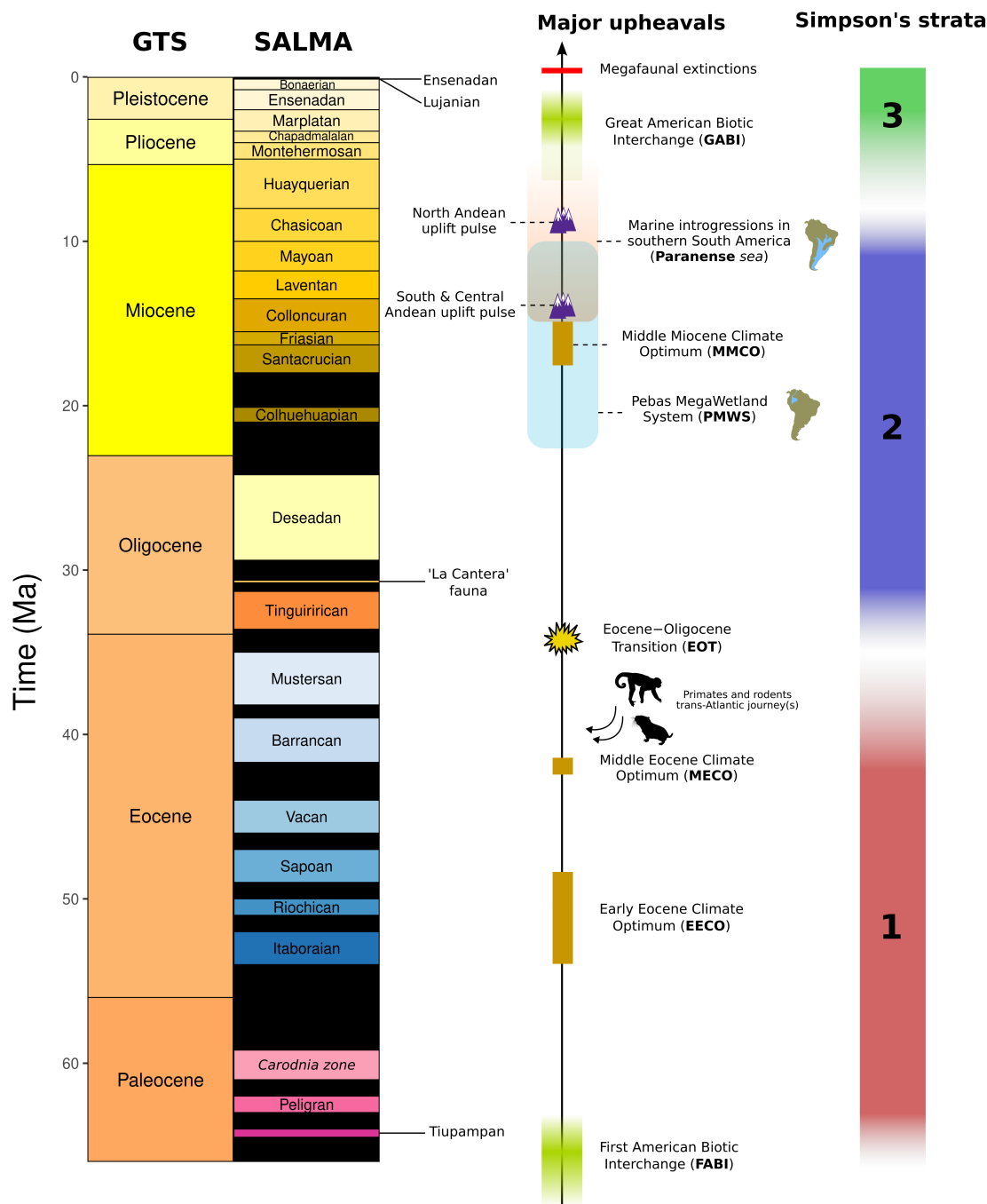


Figure 1.10: Integrative view of the Cenozoic of South America from the land mammal perspective. The South American Land Mammal Age (SALMA) scale used all over this thesis is shown, along with the corresponding Geological Time Scale (GTS) epochs, from the [International Chronostratigraphic Chart](#) (v2024/12) (Holocene not depicted). Boundaries between SALMAs were defined on the basis of the most up-to-date literature (*e.g.*, [Bellinzoni et al. 2025](#); [Dunn et al. 2013](#); [Gelfo et al. 2009](#); [Woodburne et al. 2014](#)), but note that many intervals lack absolute temporal calibration. Black spaces between SALMAs represent temporal gaps. Alongside are depicted several key biotic or abiotic upheaval in the Cenozoic history of South American mammals, as well as Simpson's three faunal strata ([Simpson 1980](#)). Silhouettes are from Phylopic (credit: Sarah Werning and Kai R. Caspar).

The most notable morphological innovation that could be mentioned is probably the increase in cheek teeth crown height (*i.e.*, hypsodonty), occurring independently among several (at least six) groups of native ungulates (particularly notoungulates) from the middle Eocene onwards (Croft *et al.* 2020). The evolution towards hypsodonty in South America is particularly interesting as hypsodont cheek teeth started to be recorded 20 million years before the first reports of hypsodonty outside South America (García-López *et al.* 2019). As they diversified, freshly-arrived caviomorph rodents (see later in this section) also, for some of them, rapidly acquired hypsodont cheek teeth, again, way before any other place in the world (late Oligocene–Early Miocene), as exemplified by the superfamily Chinchilloidea – see Chapter 2. The Eocene emergence of hypsodont cheek teeth and their Oligocene spread across lineages has long puzzled South American mammal palaeontologists (*e.g.*, Billet *et al.* 2009; Croft *et al.* 2003; Reguero *et al.* 2008, 2010). Hypsodont cheek teeth are indeed often considered as an indicator of an abrasive diet, typically, incorporating grasses, and thus, have often been related to landscape opening (Stebbins 1981). However, palaeoenvironmental evidence failed to support any concomitant grassland spreading in the regions where the first hypsodont lineages were recorded (*e.g.*, Kohn *et al.* 2004, 2015; Strömberg *et al.* 2013). Now, a consensus emerged in the community, as several lines of evidence instead suggested that the increase in cheek teeth crown height may have instead resulted from grit or volcanic dust consumption (Armella & Croft 2026; Damuth & Janis 2011; Jardine *et al.* 2012; Selkin *et al.* 2015).

Besides, by analysing two early Oligocene metatherian communities from Southern South America (one of Tinguirican age, the other being the La Cantera fauna, see Figure 1.10), Goin (2010) reported an important taxonomic turnover occurring among metatherians in the few million years that followed the Eocene–Oligocene transition (EOT). This geological boundary, that will be further explored in Chapter 1, has been identified as the most intense Cenozoic climatic disruption, as global temperature cooled down by several degrees in a few hundred thousands of years (Coxall & Pearson 2007; Liu *et al.* 2009; Westerhold *et al.* 2020). The EOT was related to a major faunal turnover among mammalian communities from the Northern hemisphere (Meng & McKenna 1998; Stehlin 1909; Stucky 2014). As an echo to the french expression ‘Grande Coupure’ (Stehlin 1909) used to characterise the mammalian turnover that occurred among European mammal assemblages, Goin (2010) thus derived the expression ‘Bisagra Patagónica’ (*Patagonian hinge*) to qualify the faunal turnover that they characterised.

Next, as we briefly evoked, Simpson’s second stratum is also characterised by the abrupt apparition of hystricognathous rodents and anthropoid primates in the South American fossil record (Antoine *et al.* 2012; Bond *et al.* 2015; Marivaux *et al.* 2023; Seiffert *et al.* 2020). Explaining such an emergence has been one of the most important challenges for South American mammal palaeontology, given that, contrary to xenarthrans (see section of stratum 1), it made no doubt that these groups were the result of a much more recent faunal supply (Simpson 1950). After several decades of intense debate, combined morphological (Hoffstetter 1972) and molecular (*e.g.*, Poux *et al.* 2006) evidence provided strong support for the “rafting hypothesis”, in which both groups would descend from Afro-Asian ancestors that would have crossed the Atlantic in the Palaeogene (*e.g.*, Antoine *et al.* 2012; Marivaux *et al.* 2023) (Figure 1.10).

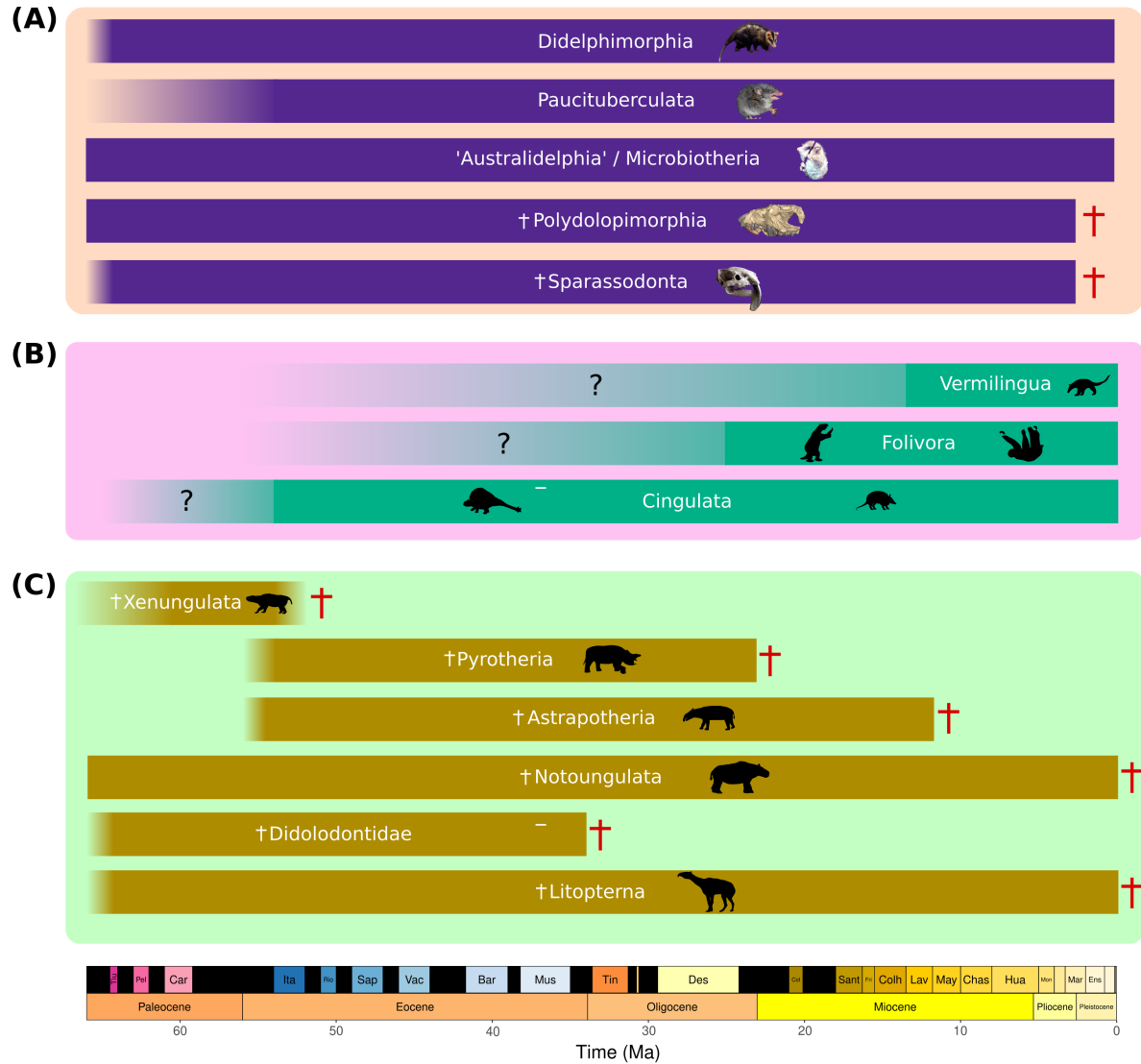


Figure 1.11: Higher-level persistence of South American mammals from Simpson's first stratum in the fossil record. **(A)** Metatherians, at the order level. Fading colours mean that the group has no known fossil record for the period, but should already exist. Photos were taken from iNaturalist (credit: Jorge Brito, Eduardo Silva Rodriguez, Thiago Chagas). *Wamradolops telloi* cranium comes from Stutz *et al.* (2026), reproduced with permission. The picture of the *Thylacosmilus atrox* cranium was taken by Claire Houck. **(B)** Xenarthra. Fading colours with question marks depict the 'xenarthran mystery', *i.e.*, that the group in question has no known fossil record for the period, but should already exist for the highlighted interval. Divergence ages between the main lineages were taken from Tejada *et al.* (2024). Vermilingua and Folivora altogether form the order Pilosa. Silhouettes are from Phylopic (credit: Julián Bayona and Sarah Werning). **(C)** South American Native Ungulates (SANUs), adapted from Croft *et al.* (2020). Fading colours depict uncertainty in the age of earliest fossil. Silhouettes are from Phylopic (credit: Julián Bayona).

Despite its wide acceptance, the timing of this fascinating trans-Atlantic journey remains under debate. Seiffert *et al.* (2020) proposed that the global sea-water drop that followed the peri-EOT climate cooling, by reducing the distance between Africa and South America, could have increased the likelihood for a trans-oceanic dispersal (assuming the latter to be inversely proportional to the distance between the source and the receiving areas). Such a hypothesis finds support in that, according to a study conducted by the same authors, the oldest fossil evidence for rodents and primates in South America cannot be older than the early Oligocene (Campbell *et al.* 2021). Conversely, Marivaux *et al.* (2023) proposed that this trans-Atlantic dispersal may have instead occurred 5 million years earlier, near the Middle Eocene Climate Optimum (MECO) (Bohaty *et al.* 2009). This interval may have been accompanied by intense meteorological episodes in the tropics (Mulch *et al.* 2015), increasing the likelihood of riverbank break-ups, and thus, the formation of floating islands serving as a means for the large-scale dispersal of small-sized mammals, such as rodents or primates (Houle 1998). This hypothesis more adequately explains the high taxonomic diversity displayed by several known early Oligocene rodent communities, accounting for taxa clearly identified as crown caviomorphs, with recognised members of Chinchilloidea (*e.g.*, Antoine *et al.* 2021; Arnal *et al.* 2022 *though some assemblages listed here may be of latest Eocene age*; Bertrand *et al.* 2012; Boivin *et al.* 2018; 2019). Such a diversity would be hard to achieve in the few hundred thousand years separating these assemblages from the EOT. Note that, given the paucity of the primate fossil record in the Palaeogene of South America – only four specimens assigned to three distantly-related lineages that notably, and contrary to rodents, are not recognised as stem representatives of the extant South American primate radiation, namely, Platyrrhini (Marivaux *et al.* 2023) – it is currently impossible to apply such a limitation to the primate scenario proposed by Seiffert *et al.* (2020).

In addition, the early Oligocene interval has been recognised as the most likely time frame for the colonisation of the West Indies by several taxa, including crown caviomorph rodents (Marivaux *et al.* 2020b), sloths (Tejada *et al.* 2024) as well as other terrestrial taxa (Blackburn *et al.* 2020; Chamberland *et al.* 2018). These colonisations would have been favoured by the emergence of an ephemeral land bridge between the Greater Antilles and the South American landmass – the Aves Ridge –, following the aforementioned sea-level drop after the EOT, a hypothesis called GAARlandia (Cornée *et al.* 2021; Iturralde-Vinent & MacPhee 1999; Philippon *et al.* 2020). However, it should be mentioned that such a hypothesis has raised criticism (*e.g.*, Ali 2012; Ali & Vences 2019).

Simpson's second stratum persisted through the Oligocene and the Miocene. Keystone late Oligocene localities, such as Salla (Bolivia) or the Deseado massif (Argentina) (Figure 1.9) provided unprecedented insight into South American mammal diversity at that time. In particular, they helped identify the first records of sloths (Pujos & De Iuliis 2007) as well as to better constraint the diversification of Notoungulata, Litopterna, Astrapotheria and the extinction of Pyrotheria near the Oligocene–Miocene (Croft *et al.* 2020) (Figure 1.11 C). Then, from the early Miocene, sampling gets increasingly less temporally discontinuous (Figure 1.10), thus better constraining our knowledge of the evolution of mammals on the continent. In addition, molecular and fossil data agree that most of the crown extant South American mammal families that survived until the present diverged in the Miocene (Álvarez-Carretero *et al.* 2022). Thus, mammal diversity from the Late Oligocene to the Late Miocene was flourishing, and many invaluable sites or geological formations (*e.g.*, Santa Cruz Formation, La Venta, Quebrada Honda,... see Figure 1.9 as well as Croft (2016)) yielded precious discoveries attesting that statement.

Third stratum: Inter-american exchange, massive faunal disruptions, and establishment of the extant mammal biodiversity (latest Miocene–Present)

Please note that, all-over this manuscript, “North America” will actually refer to North and Central America, roughly, everything above the Isthmus of Panama.

The flourishing diversity that characterised Simpson’s second stratum was massively disrupted in the late Neogene. As seen in Section 1.4.1, ca. 3 Ma, a sustainable landbridge connecting the Americas emerged, known as the Isthmus of Panama. Though noticeable examples of earlier colonisations have been documented in the fossil record (Woodburne 2010), it is consensual to consider that the closure of the isthmus of Panama catalysed the admixture between North and South America’s endemic mammalian faunas, an event famously known as the ‘Great American Biotic Interchange’ (GABI) (Webb 1985) (Figure 1.10).

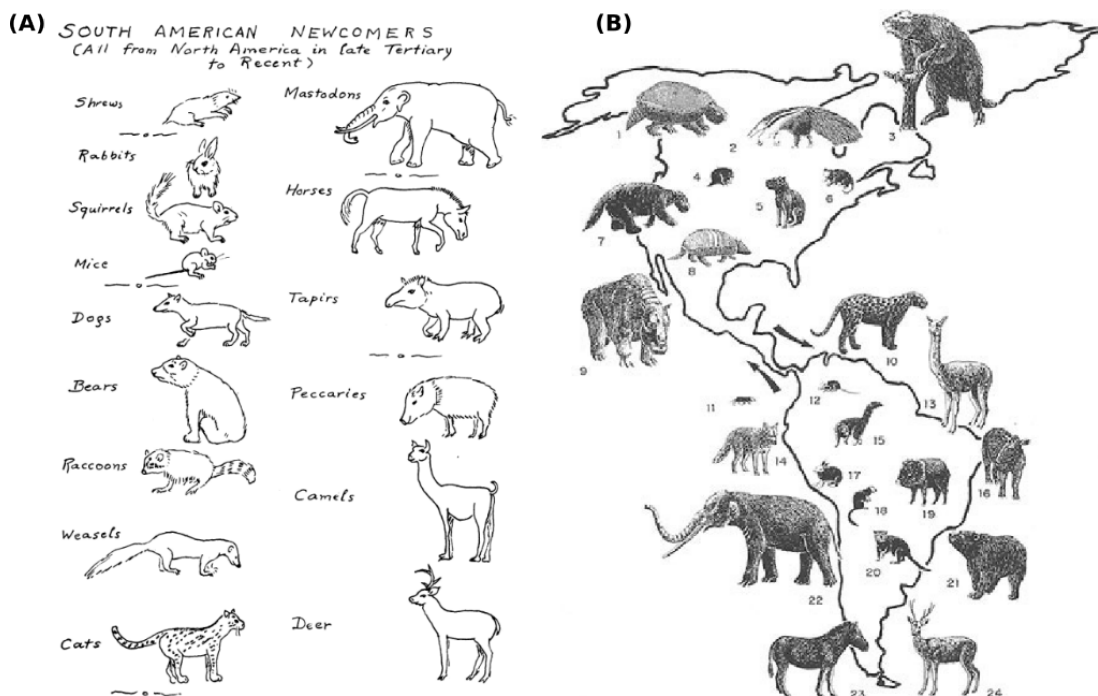


Figure 1.12: (A) Hand-drawing of the North American taxa recorded in the fossil record of South America, hence considered ‘newcomers’, made by Simpson (1950). (B) Schematic interpretation of land mammalian exchange across the Americas following the GABI, from (Vrba 1992). All the taxa shown in North America are of South American origin, and vice-versa. *In North America*: 1. glyptodon, 2. anteater, 3. and 7. ground sloths, 4. New-world porcupine, 5. capybara, 6. opossum, 8. armadillo, 9. toxodontid notoungulate. *In South America*: 11. shrew, 12. sigmodontine cricetid, 13. camelid, 14. canid, 15. mustelid, 16. tapir, 17. rabbit, 18. squirrel, 19. peccary, 20. racoon, 21. bear, 22. gomphotheriid, 23. horse, 24. cervid.

As described by Woodburne (2010), although the early phases of the GABI were characterised by roughly symmetrical exchanges (as illustrated by the various groups on Figure 1.12 B), during the Pleistocene, the fossil record suggests that an asymmetry of exchange gradually established, with North American taxa colonising South America with a higher success than the other way around.

At present, it is estimated that around 50% of South American mammal's generic diversity descend from a North American lineage, whereas this proportion falls down to some 10% when considering genera with a South American ascent living in North America (Webb 1991). The latter proportion is considerably reduced to a few single species when excluding tropical North America, a region that even Wallace recognised as “the portion of the North American continent which belongs zoologically to South America” (Wallace 1876). Consequently, it stands out that the GABI has been highly asymmetric.

What shaped the striking asymmetry of this lineage exchange has been one of the most puzzling issues in South American mammal palaeontology. Several hypotheses have indeed been proposed to explain such a exchange imbalance, that will be intensively reviewed and discussed in Chapter 3. The first hypotheses formulated involved some kind of competitive superiority of North American taxa out of South Americans (*e.g.*, Simpson 1950). Later hypotheses instead placed the emphasis on the role of palaeoclimate change (*e.g.*, Vrba 1992; Webb 1991). Simpson (1980) even termed such an exchange a ‘fortuitous experiment’, arguing that if things were to be repeated, the resulting pattern would thus not necessarily be equivalent. On top of this, confounding effects may arise from megafaunal extinctions (Hadler *et al.* 2026; Svenning *et al.* 2024), several studies arguing that hunting may also have caused the demise of endemic (large-sized) south American clades in South America (Boscaini *et al.* 2025; Prates & Perez 2021). Consequently, despite being the subject of a wide range of studies, the drivers behind the asymmetry of the GABI remain an open question, and one of the core aims of the present doctoral work is to contribute to a better understanding of this fascinating enigma.

1.4.3 Challenges for South American mammal palaeontology

Despite being unique and invaluable, several challenges arise as we intend to work with the South American mammal fossil record, in addition to those previously mentioned arising while using fossils (Section 1.1.4).

As it shows on the maps displayed in Figure 1.9, the fossil record of South American mammals is strikingly uneven across space, with the persistence of a huge gap merely overlapping with the plain of the Amazon river, the Brazilian Highlands and the Guianian Shield (Figure 1.8). Almost 30 years ago, Flynn & Wyss (1998) already pointed the need to fill the pronounced gap in the tropical fossil record of South American mammals, being highly sparse, with the notable exception of the exceptional Miocene fauna from the La Venta locality, Colombia (Carrillo *et al.* 2023) (Figure 1.9). This is primarily the result from three factors. First, except along the Andes where the subduction has actively promoted sediment accumulation in the Andean foreland basin (Figure 1.8), tropical South American soils experience little subsidence, and consequently, being less dense than continental crust, hardly can sediments get buried in the Amazonian plain (and thus offering the possibility for fossils to form). Second, tropical soils are markedly more active in terms of microfauna. Decomposition there is thus more efficient than outside the tropics, reducing the probability for a dead body to fossilise. Last, obvious accessibility issues hamper fossil-hunting in the tropical rainforest. Without invoking geopolitical factors, low-latitude fossil-bearing localities from Western Amazonia are indeed almost all located nearby brand-new roads across the forest or along rivers, posing additional logistic issues (Figure 1.13).

Substantial improvements should however be stressed, notably in the Peruvian Amazonia, where exceptional fossil deposits have been found (*reviewed in* Antoine *et al.* 2017). These are greatly exemplified by the regions of Contamana and Shapaja, where nearly 100 different vertebrate-bearing localities spanning distinct intervals of the Cenozoic have been characterised (Antoine *et al.* 2016;

Antoine *et al.* 2021), including CTA-27, where the oldest caviomorph rodent community of the continent was unearthed (Antoine *et al.* 2012). Significant contributions should also be mentioned from the early Oligocene locality of Santa Rosa (Campbell *et al.* 2004), which yielded two out of the three basalmost primates found in South America to date (Bond *et al.* 2015; Seiffert *et al.* 2020). Despite all these costly efforts, critical time intervals (such as the middle-late Oligocene or the early Miocene) remain unsampled under tropical latitudes, superimposing on the existing SALMA scale’s gaps (depicted in black on Figure 1.10). Consequently, every palaeontologist who ever worked in tropical South America knows this ‘South American mammal fossil record paradox’, in which the region encompassing the greatest diversity at the scale of the continent (and even the world, see Figure 1.2) is also, by far, the most fossil-poor.

Such an incompleteness was shown to significantly affect upstream macroevolutionary analyses relying on the South American mammal fossil record (Ugarte *et al.* 2025), and should thus be kept in mind throughout the rest of this dissertation.



Figure 1.13: Example of a fossil locality along the Tambo river, Peruvian Amazonia (photo by the author).

1.5 Objectives of this doctoral work

This thesis aims to improve the understanding of key moments in South American mammal evolutionary history through an interdisciplinary approach of macroevolution, macroecology and palaeobiology.

First, emphasis will be placed on the Eocene–Oligocene interval (*ca.* 56–23 Ma), genuinely straddling Simpson’s first and second strata, and encompassing the most dramatic Cenozoic climate disruption (the Eocene–Oligocene Transition, EOT, *ca.* 34 Ma). Though the EOT has been associated to a marked mammal turnover (sometimes even considered as a mass extinction) in the Northern Hemisphere, continent-scale evidence in South America is lacking. The first chapter of this thesis (Chapter 1) will contribute to the discussion on the status of the EOT as a mass extinction by reconstructing the Eocene–Oligocene diversification dynamics of South American mammals, relying on an unprecedented dataset and cutting-edge preservation-aware methods.

Throughout the course of the first chapter, long-standing issues about diversification regimes under the tropics raised since the times of Wallace (Section 1.2.2) will be approached. Specifically, whether tropical and extratropical clades exhibit similar or distinct macroevolutionary trajectories will be explored, along with the latitudinal (un)conspicuousness of the environmental forces underlying the reconstructed diversification trajectories.

Next, this thesis will also contribute to a better understanding of the emergence of singular phenotypic features among South American land mammals. By reconstructing the trait- and environment-dependent diversification history of Chinchilloidea (Rodentia: Caviomorpha), a rodent clade poorly-diverse at the present with a rich fossil record encompassing the largest rodents to ever fool the ground of the Earth, foundational hypotheses on trait-dependent diversification (Section 1.2.1) will be explored. Ultimately some light will be shed on the existence of differential macroevolutionary tempo between chinchilloid body size and hypsodonty categories, meanwhile pinpointing the likely environmental drivers underlying the diversification dynamics of this emblematic clade. Also, the first chapter of this thesis will provide quantitative insights into how grassland spreading shaped South American mammal diversification dynamics in the Palaeogene, which has historically been invoked as the driver behind the strikingly earlier emergence of hypsodonty in mammals from South America compared to the rest of the world, but further largely criticised.

Also, in its last chapter, this thesis will provide valuable mechanistic insights into what shaped a textbook example of asymmetric lineage exchange following a geodispersal event, namely, the Great American Biotic Interchange (GABI) (Section 1.4.2). The imbalanced success of North American mammal lineages compared to South Americans in the aftermath of the closure of the isthmus of Panama has indeed yielded to a wide range of hypotheses. By means of computer experiments, this thesis will specifically assess whether such an asymmetry can be linked to palaeoclimate dynamics, a long-considered paradigm.

The findings produced over this thesis will ultimately be replaced in a broader framework, discussing the likely triggers of macroevolutionary change among South American mammals, and several perspectives will be provided to enhance further macro-eco-evolutionary reflexion on the fate of this fascinating study model.

Chapter 1. The fate of South America's endemic mammalian fauna in response to the most dramatic Cenozoic climate disruption

Foreword

Understanding what drove the emergence and demise of lineages during biodiversity crises may provide key insights at the time of the sixth mass extinction. Here, using a vetted fossil database, we explored how South American mammals (SAM) responded to the most dramatic Cenozoic climate cooling episode, the Eocene–Oligocene transition (ca. 34 Ma). We found that SAM experienced a gradual and long-standing diversity decline strongly related to late Eocene progressive climate cooling, followed by a major taxonomic turnover in the Oligocene, driven by diversity-dependent factors, Andean uplift and perhaps grassland expansion. SAM were therefore more subject to long-term diversity changes rather than abrupt ones, contrary to what would be expected in a mass extinction scenario.

The content of this chapter was published in an international peer-reviewed journal ([Buffan *et al.* 2025](#)).

Abstract

Around 34 million years ago (Ma), the Eocene–Oligocene transition (EOT) marked the most dramatic global climatic cooling of the Cenozoic. On a planetary scale, palaeontological evidence suggests that this transition was associated with major faunal turnovers, sometimes even regarded as a mass extinction crisis. In South America, there is no consensus on the response of the endemic mammals to this transition. Here, using a vetted fossil dataset and cutting-edge Bayesian methods, we analyzed the dynamics of South American mammal (SAM) diversification and their possible drivers across latitude (tropical vs. extratropical), taxonomic groups, and trophic guilds throughout the Eocene–Oligocene (ca. 56–23 Ma). Our results did not evidence any mass extinction among SAM at the EOT. Instead, they experienced a gradual and long-term diversity decline from the middle Eocene to the early Oligocene, followed by a sudden waxing-and-waning diversity associated with a large taxonomic – but not ecological – turnover. Tropical and extratropical lineages have had very distinct macroevolutionary histories. No effective change in the pace at which tropical lineages diversify was found, thus favoring the tropical stability hypothesis proposed by Wallace. Diversity-dependent effects, temperature and Andean uplift were recovered as probable drivers of SAM diversification across the period. Contrasting evidence casts doubt on the common hypothesis primarily linking Oligocene faunal changes to grassland expansion. Our findings illustrate the uniqueness of the deep-time interplay between endemic SAM and their physical environment in a context of climatic shift, highlighting the need to consider regional idiosyncrasies for understanding the coevolution of life and climate.

Introduction

Today, South America is home to a unique mammalian fauna, hosting one of the most species-rich regions of the world, Amazonia (Willig *et al.* 2003). Although the extant mammalian taxonomic diversity in the continent is astonishing, it is a poor reflection of what can be drawn from the fossil record. In fact, throughout most of the Cenozoic (i.e., the last 66 million years), South America was isolated from other landmasses (Scotese 2021). In this insular context, terrestrial mammal lineages that inhabited this huge landmass experienced unique radiations (Beck *et al.* 2023; Boivin *et al.* 2019; Croft *et al.* 2020; Goin *et al.* 2016; Tejada *et al.* 2024). According to Simpson (1980), South American mammal (SAM) communities through the Cenozoic are well summarized by three successive ‘faunal strata’. The first one, stretching from the early palaeocene (66 million years ago, Ma) to the late Eocene (ca. 37 Ma), is characterized by the early establishment and radiation of endemic lineages, mostly xenarthrans, metatherians and native ungulates. Then, from the early Oligocene (ca. 33 Ma) until the Late Miocene-Early Pliocene (ca. 5 Ma), key morphological innovations appeared among native lineages, leading to a so-called ‘faunal modernization’ (Croft *et al.* 2008). In the meantime, rodents and primates, Afro-Asian lineages that most likely reached South America in the middle-late Eocene (Antoine *et al.* 2012; Bond *et al.* 2015; Marivaux *et al.* 2023), diversified locally (Beck *et al.* 2023; Boivin *et al.* 2019). In Plio-Pleistocene times, this endemic fauna was massively disrupted. Meanwhile, the closure of the Panama isthmus (Montes *et al.* 2015) had paved the way for major faunal exchanges between North and South America, described as the ‘Great American Biotic Interchange’ (Marshall 1988), and immigrant taxa from North America (e.g., carnivores, perissodactyls, artiodactyls, or proboscideans) appeared in the South American fossil record. In the last millions of years, many endemic South American lineages declined, or even went totally extinct (e.g., litopterns, notoungulates, sparassodonts, glyptodonts, ground sloths).

Close to the limit between Simpson’s first and second faunal strata, the Eocene–Oligocene transition (EOT, ca. 33.9 Ma) marked the most dramatic Cenozoic climatic change episode worldwide (Coxall & Pearson 2007). At that time, simultaneously, global climate massively cooled down (Westerhold *et al.* 2020), ocean level dropped (Miller *et al.* 2020), and the whole Antarctic continent turned into a giant ice sheet (Liu *et al.* 2009). The prevailing mechanism proposed to explain these major environmental changes is the onset of the Antarctic circumpolar current (Toumoulin *et al.* 2020). Concomitantly, a major biotic turnover among mammals, known as the ‘*Grande Coupure*’, occurred in Europe (Stehlin 1909). This event, punctuated by the arrival of immigrant taxa from Asia, saw the demise of many endemic European lineages (Escarguel *et al.* 2008; Legendre *et al.* 1991; Weppe *et al.* 2023). Although European-oriented case studies dominate the literature about the topic, similar disruptions among mammalian faunas occurred at the same time in other regions of the world, such as Asia (Kraatz & Geisler 2010; Meng & McKenna 1998; Sun *et al.* 2014), North America (Figueirido *et al.* 2012; Stucky 2014), Africa (Vries *et al.* 2021) or to some extent Madagascar (Godfrey *et al.* 2020). Recently, Hoyal Cuthill *et al.* (2020) even came to rank this geological boundary as the second-most important extinction event throughout the history of life.

In central and southern South America, substantial changes have been documented in metatherians (Abello *et al.* 2020; Goin 2010) and native ungulates (Croft *et al.* 2020; Solórzano *et al.* 2024) near the EOT. As an echo to the ‘*Grande Coupure*’, Goin (2010) even introduced the term ‘*Bisagra Patagónica*’ (Patagonian Hinge) to characterize the changes among metatherian communities in Patagonia possibly related to the EOT. General morphological trends have been highlighted in ‘modernised’ notoungulate faunas, with a net increase in body mass and cheek teeth’s crown height (Solórzano & Núñez-Flores 2021). The latter changes, by analogy with the evolutionary fate of extant groups, suggested the

emergence of the grazing feeding mode, and was postulated to result from the expansion of grasslands, promoted by post-EOT climate cooling and aridification (Reguero *et al.* 2010; Solórzano *et al.* 2024; Stebbins 1981). In northern South America, changes in methaterian and rodent communities around the EOT have also been suggested despite the paucity of known Palaeogene low-latitude fossil localities to date (Antoine *et al.* 2021). Additionally, geological, palaeontological and phylogenetic evidence suggests that the Eocene–Oligocene was a key interval for the dispersal of SAM toward Caribbean Islands, further playing a central role in the architecture of these unique insular ecosystems (Cornée *et al.* 2021; Marivaux *et al.* 2020b; Philippon *et al.* 2020; Tejada *et al.* 2024).

Despite general agreement that a faunal remodelling occurred in SAM communities near the EOT, the evolutionary tempo and mode remain unknown. At macroevolutionary scales, as we would expect in a mass extinction scenario (*SI Appendix, Supplementary text*), it is unclear whether this faunal renewal was characterized by extinction of Eocene lineages followed by diversification of the surviving ones (Guinot & Condamine 2023), or whether lineages gradually transitioned until they became ‘modernised’. A crucial question underlying these conflicting macroevolutionary hypotheses is whether or not extinction occurred in the tropics, and at what level. Indeed, it has long been postulated that the climatic stability of tropics would make tropical lineages less vulnerable to extinction than extratropical lineages (Wallace 1878). Also, the ecological selectivity and the likely drivers of this apparent faunal transition have remained underexplored. For instance, there is currently no consensus regarding the role of temperature (Kohn *et al.* 2004, 2015) or grassland expansion (Strömberg *et al.* 2013) in shaping the macroevolutionary fate of SAM. Finally, there is a lack of integrative frameworks at the continental scale, with most case studies at the local scale, mostly in southern South America. Nevertheless, the Palaeogene mammal fossil record in South America is spatially and temporally uneven. Spatial gaps cover large areas, such as most of the Amazon basin, and important time intervals have remained unsampled to date (*SI Appendix, Figure S1 and S3*). Accordingly, it has been historically challenging to address questions regarding the history of SAM at the continental scale.

Here, we aim to fill these gaps by examining the available fossil record of SAM. We assembled a genus-level fossil dataset consisting of 3,378 occurrences (distributed in 1,115 species, 536 genera, 96 families, and 18 orders) spanning from the late palaeocene to the Early Miocene. Relying on a fossil-based Bayesian birth-death framework accounting for variable preservation rates through time and across taxa (Silvestro *et al.* 2014b), we estimated the tempo and modes of the mammalian diversification dynamics in South America throughout the Eocene–Oligocene interval (*ca.* 56–23 Ma). We then disentangled the underlying taxonomic, functional, and geographic (tropical vs. extratropical) bases of the inferred continent-scale patterns of diversity and diversification among SAM faunas. We finally quantified the most likely biotic and abiotic drivers explaining their diversification dynamics.

Results and discussion

No mass extinction: South American mammal diversity gradually declined well before the EOT

We analyzed the fossil database (Table S1) both at the genus and species levels using the process-based Bayesian framework of diversification accounting for varying preservation implemented in PyRate (Silvestro *et al.* 2014b). Our results highlight that the overall mammal diversity gradually declined from the end of the early Eocene – shortly after the Early Eocene Climate Optimum (EECO, 45) – to the end of the early Oligocene (decline by respectively 34% and 52% in the number of genera and

species between 48 and 32 Ma), and then further increased in the late Oligocene (Figure 1.14 D and SI Appendix, Figure S8-11). No massive loss of lineages was evidenced at the EOT. This diversity trajectory results from a significant decrease in origination rate during the early Eocene (Figure 1.14 B and SI Appendix, Figure S8-11). Additionally, we recover a joint significant increase in the pace of origination and extinction (i.e., increased turnover) during the late Eocene, with the extinction rate always being higher than the origination rate (slightly negative net diversification rate; Figure 1.14 B–C and SI Appendix, S8-11). Similar diversification patterns are recovered when analyzing the dataset at the genus or species level (SI Appendix, Figure S8-11), after removing singletons (SI Appendix, Figure S12-13) or disabling several occurrences of the same lineage that co-occur temporally and geographically (SI Appendix, Figure S14-15). Moreover, general trends in diversity and diversification dynamics are conserved when working with occurrences assigned to South American Land Mammal Ages (SALMAs; SI Appendix, Figure S9 and S11) instead of sub-epochs (see Methods; SI Appendix, Figure S1-2 and Extended Material and Methods), or when considering attempts to derive early Oligocene ages to western Amazonian occurrences originally assigned to an Eocene age (SI Appendix, Figure S32).

Recently, an analysis of the global Phanerozoic fossil record classified the EOT as the second most intense mass extinction episode (Hoyal Cuthill *et al.* 2020) (see SI Appendix, Supplementary text for a definition of a mass extinction). Although this ranking was unexpectedly high, drastic extinctions at the EOT are supported by analyses of the European (Stehlin 1909) and Asian (Meng & McKenna 1998) mammal fossil record. Applying a methodological framework similar to ours, Weppe *et al.* (2023) recovered a strong extinction signal among endemic Western European artiodactyls at the EOT, with the loss of 62% of genera and 77% of species in only 1 My. Other studies working on mass extinctions illustrated similarly high levels of lineage loss for various study models, for instance in insects at the Permian/Triassic boundary (82% genus loss) (Jouault *et al.* 2022) or in elasmobranchs – sharks, skates and rays – at the Cretaceous/palaeogene boundary (44.94% genus and 62.6% species loss) (Guinot & Condamine 2023). Here, given the relatively small number of lineages lost at the Eocene–Oligocene boundary – 15 genera (19%), 19 species (22%) –, our results are not consistent with a mass extinction event among SAM at the EOT. Instead, we highlight a long-term decline in SAM diversity, from the middle Eocene onward, contrasting with the expected short temporality of a mass extinction crisis. To our knowledge, our work is the first to provide quantitative support for such a pattern in SAM. These differences between South America and other continents could come from the tropical nature of and the ecological continuum within a significant part of the landmass. Stemming from the cornerstone hypothesis of Wallace linking the striking similarities between tropical biomes from all-over the world to the uniformity of tropical climate across space and (shallow) time (Wallace 1878), tropical climate stability would act as a buffer for tropical lineages against extinction. Also, the isolated condition of the landmass has been proposed as a probable cause explaining the singularity of SAM near-EOT macroevolutionary fate, as it greatly limited the possibilities for immigration events. Yet, immigrations were hypothesized to have played a key role in EOT-related extinctions in other parts of the world (Goin *et al.* 2016) (but see Escarguel *et al.* 2008).

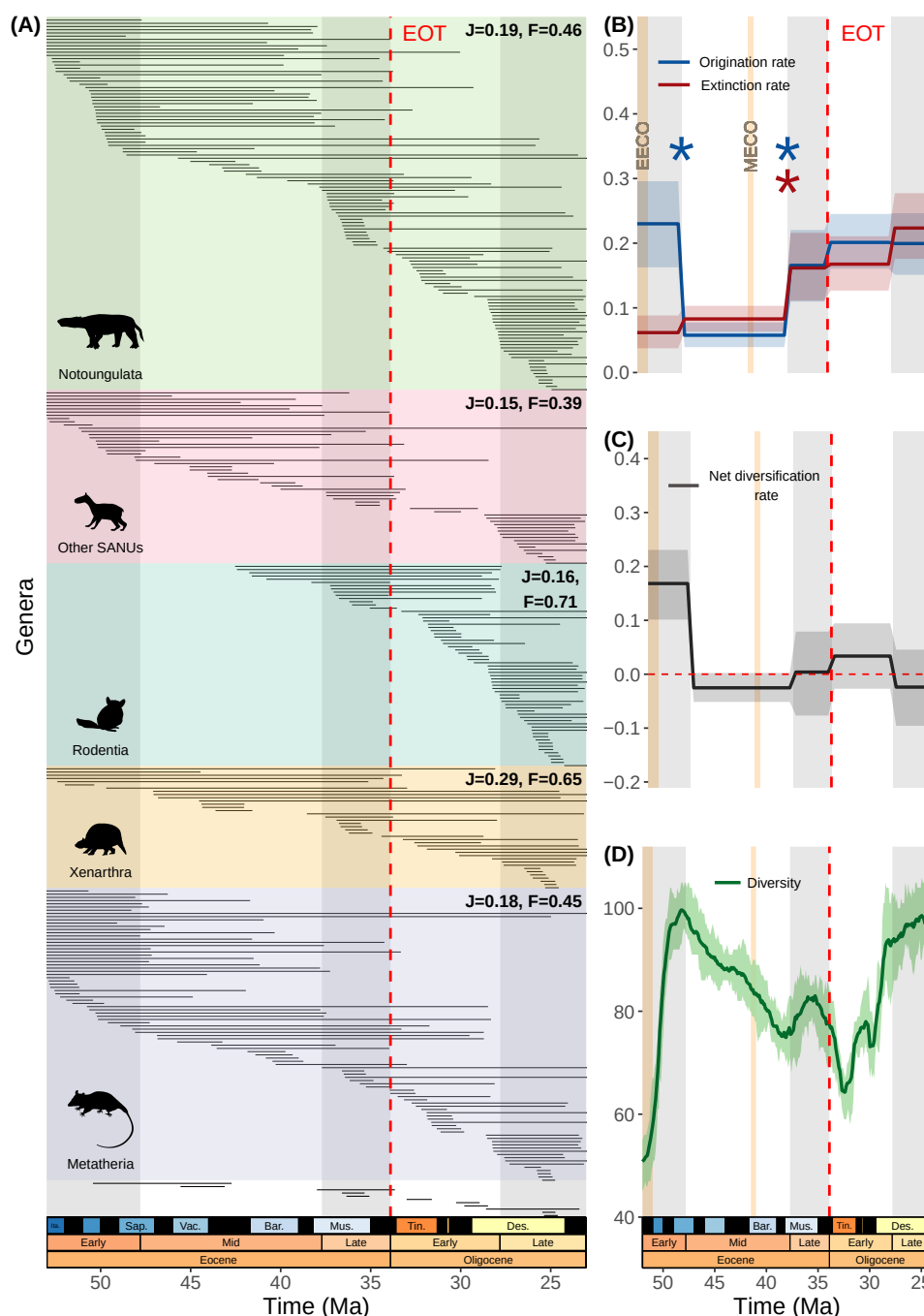


Figure 1.14: Bayesian reconstructions of taxon longevity (A), diversification (B–C) and diversity (D) dynamics of South American Mammals from the early Eocene to the late Oligocene (genus level). The plot on the left (A) shows the longevity – i.e., interval between origination and extinction times – of each Eocene–Oligocene (373 out of 529 genera) South American mammal genus estimated with PyRate. J- and F-values respectively represent Jaccard's and Forbes' similarity indices computed between Eocene and Oligocene assemblages for each of the five groups. The right part of the panel shows the origination (blue) and extinction (red) rates through time (B), the net diversification rate (origination minus extinction) (C), and diversity (D) of all South American mammals throughout the study period. Full lines represent median estimates and ribbons their associated 95% higher posterior density. Light brown vertical bands indicate two key climate events, the Early Eocene Climate Optimum (EECO, ~53–51 Ma) and the Middle Eocene Climate Optimum (MECO, ~41.5–41 Ma). Stars on the (B) plot indicate significant rate shifts. Organisms' silhouettes are from PhyloPic and were added thanks to the R package rphylopic (98). Silhouettes were contributed by Julián Bayona, 2016 (CC BY-SA 3.0), Milena Cavalcanti,

Ubiquitous taxonomic – but not ecological – turnover among the main mammalian groups across the study period

Our analyses of the overall SAM diversification dynamics failed at characterizing any mass extinction event at the time of the EOT, but recovered a late Eocene increase in mammal turnover (joint increase in origination and extinction rates), followed by an Oligocene waxing-and-waning diversity (Figure 1.14 B and D). As a result, Eocene and Oligocene SAM appear taxonomically dissimilar. Indeed, the apparent proportion of shared genera between each temporal fauna, represented by the Jaccard's similarity index, is below 20% for four out of the five main groups we studied (Figure 1.14 A and SI Appendix, Figure S38). Nevertheless, the corrected Forbes index, that was shown to perform more accurately when evaluating pairwise similarity between samples subject to data incompleteness (Alroy 2015) reveals higher levels of faunal similarities, but still below 50% for three out of our five main studied groups (Figure 1.14 A and SI Appendix, Figure S39). This does not call into question the occurrence of a taxonomic renewal, but suggests that it occurs at an attenuated magnitude than suggested by the Jaccard's index. The corrected genus longevity estimates corroborate results already established in the literature about mammal taxonomic replacement near the EOT, but also highlight previously undescribed patterns (Figure 1.14 A and SI Appendix, Figure S18-21).

For notoungulates, the most taxon-rich order of our dataset (encompassing a third of our total occurrences; SI Appendix, Table S1), our inferences corroborate a long-recognized turnover at the order scale between the late Eocene and the early Oligocene (Croft *et al.* 2008). This goes in line with a recent species-level analysis of southern South American notoungulate diversification dynamics, pointing toward an enhanced turnover in the early Oligocene, mostly among smaller-sized taxa (Solórzano *et al.* 2024). According to our findings, this turnover is particularly striking at the family level, with the synchronous extinction and origination of three (Archaeopithecidae, Notostylopidae, and Oldfieldthomasiidae) and four families (Homalodotheriidae, Hegetotheriidae, Mesotheriidae, and Toxodontidae), respectively (SI Appendix, Figure S19-20). Most of these newly-arising families, in addition to two surviving ones (Archaeohyracidae and Interatheriidae), convergently evolved larger body sizes and higher teeth crown height (hypsodonty) than their basal counterparts, mostly during the Oligocene (though the first records of a hypsodont notoungulate lineage dates back to the middle Eocene, 20 million years before the first record of hypsodonty outside South America) (Solórzano & Núñez-Flores 2021; Stebbins 1981). As notoungulates likely represented a huge proportion of herbivorous lineages, these innovations likely implied changes in the architecture of the primary feeder mammalian guild between the Eocene and the Oligocene ecosystems at continent scale.

Considering the other native ungulate orders (i.e., Litopterna, Astrapotheria, and Pyrotheria), apart from the late Eocene extinction of Trigonostylopidae (Astrapotheria), evidence for higher-level taxonomic changes near the EOT is less obvious (SI Appendix, Figure S23). However, there seems to be a strong intra-family taxonomic renewal among most litopterns (i.e., Proterotheriidae, Macrauchenidae, and Adiantidae) and astrapotheriids (Astrapotheria) between the Eocene and Oligocene. These groups have been affected by a diversity decline from the late Eocene to the early Oligocene (SI Appendix, Figure S16D-F and S17D-F). Given that, just like notoungulates, these groups likely had herbivorous diets, the highlighted changes also surely affected the architecture of the ecosystems between the Eocene and the Oligocene.

In the case of metatherians (SI Appendix, Figure S18), our results are in line with previous findings about their taxonomic renewal near the EOT (Goin *et al.* 2016). On one hand, entire superfamilies or sub-orders like Bonapartharioidea or Polydolopiformes (both subclades of the order Polydolopimorphia according to Goin *et al.* (2016)) went extinct during the early Oligocene, meanwhile others, like

Argyrolagoidea (also a subclade of Polydolopimorphia according to Goin *et al.* (2016)), palaeothentoidea (Paucituberculata) and Hathliacynidae (Sparassodonta) originated and/or diversified. We may specify that the superfamily Palaeothentoidea appeared in the middle-late Eocene with the two stem genera Sasawatu and Perulestes, but the Palaeotenthidae and Abderitidae families diversified from the early Oligocene onward. On the other hand, intra-clade taxonomic replacement occurred among Borhyaenoidea (Sparassodonta) and Microbiotheria. As mentioned before, Goin (2010) established that the most dramatic turnover among metatherian faunas occurred near the EOT (the ‘Bisagra Patagónica’). Our corrected genus age reconstructions clearly bring qualitative support to such a macroevolutionary fate. However, when estimating metatherian diversification dynamics, no significant rate shift was found, and the overall number of genera through time remained stable (SI Appendix, Figure S17M-O). Quantitatively, our diversification analyses supported a constant turnover rate across the Eocene–Oligocene interval, which is likely a consequence of the low representation of metatherians in our dataset (219 occurrences, 54 genera).

As for cingulates, representing the majority of our xenarthran occurrences, we highlight a slowdown in Dasypodinae diversity, and the origin of Euphractinae in the middle-late Eocene (SI Appendix, Figure S21), in agreement with Ciancio *et al.* (2013). Just like for metatherians, our diversification reconstructions provided support for a constant-rate diversification throughout the period of interest, with a stable number of genera (SI Appendix, Figure S16I-L and 17I-L).

Regarding caviomorph rodents, the oldest representatives of the clade are found to originate in the end of the Eocene (Figure 1.14 A and SI Appendix, Figure S22). Then, the group diversified at a constant rate until it reached a mid-Oligocene peak in extinction rate (SI Appendix, Figure S16G-I and S17G-I), and no main taxonomic replacement near the EOT is to be noticed.

To track the evolution of ecological composition between the Eocene and Oligocene SAM, we summarized trophic palaeocommunities by four dental ecomorphotype-derived categories: herbivorous, carnivorous, insectivorous and omnivorous. We used the origination and extinction ages estimated by PyRate to assess the number of genera through time within each ecological category and further computed the relative proportion of each trophic category at the geological sub-epoch level (SI Appendix, Figure S1 and Extended Materials and Methods). Our findings illustrate that the structure of trophic palaeocommunities remained stable throughout the entire time interval (Figure 1.15). We report a slight increase in the relative proportion of herbivorous lineages in assemblages of decreasing age, at the expense of omnivorous and insectivorous. In addition, ecological dissimilarity increases with the age difference between two assemblages, but remains low (below 25%), especially between the late Eocene and early Oligocene assemblages, when it equals 6.1% (Figure 1.15).

Our results provide support for a taxonomic renewal between Eocene and Oligocene SAM faunas, with few changes in the architecture of ecosystems, thereby suggesting intra-guild renewal. Interestingly, the onset of this faunal turnover temporally matches the timing of ‘the most fundamental biotic division in the Cenozoic’ in the marine realm, that is, between the middle and the late Eocene (Prothero 1994).

Latitudinal heterogeneity in diversification histories

Since the late break-up of the Gondwana supercontinent ca. 130 Mya, the South American landmass has offered a duality between tropical and non-tropical habitats (Scotese 2021). This duality is particularly interesting as it allows for testing a long-standing question in macroevolution as to whether diversification processes occurred the same way inside and outside the tropics (Wallace 1878). Based

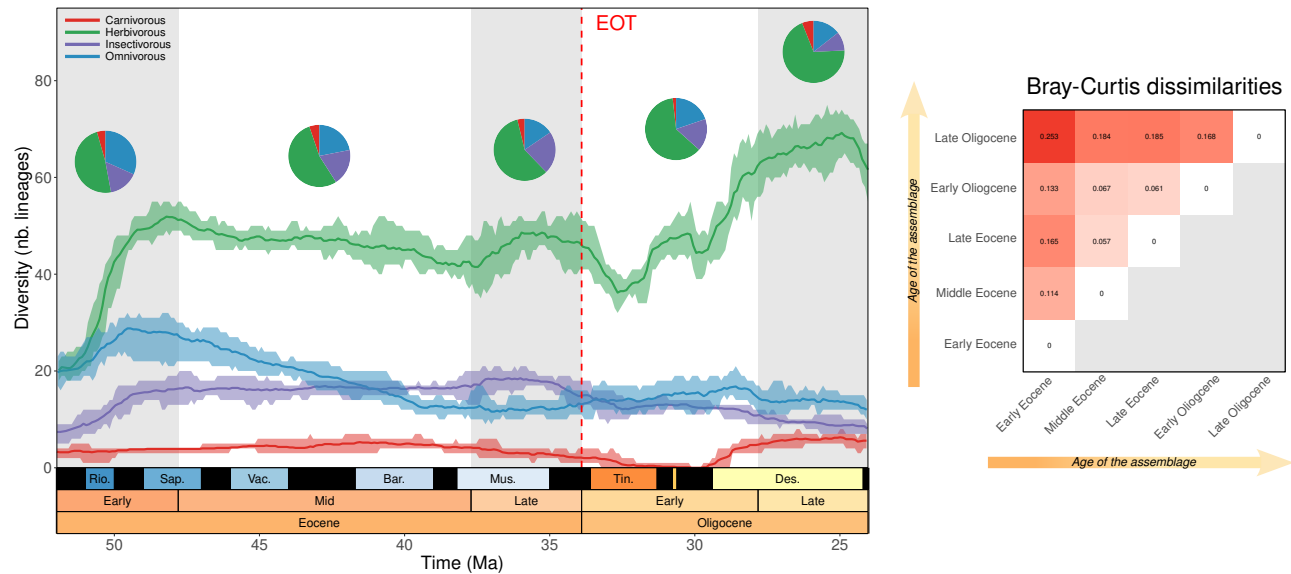


Figure 1.15: Evolution of genus-level functional assemblages between the Eocene and the Oligocene. The plot on the left shows the median (full line) and 95% HPD (ribbon) of estimated genus richness among the four trophic classes we defined based on dental ecomorphotypes: carnivorous (red), herbivorous (green), insectivorous (purple) and omnivorous (blue). Pie charts indicate the relative abundance of each trophic group for each sub-epoch. The matrix on the right shows pairwise Bray-Curtis dissimilarities between sub-epoch-level trophic communities, with darker colour indicating higher dissimilarities. Geological and SALMA timescales were added thanks to the *deptime* R package (Gearty 2023). Abbreviations for SALMA. Des: Deseadan, Ting: Tinguirirican, Must: Mustersan, Bar: Barrancan, Sap: “Sapoan Fauna”, Rio: Riochican. The small band between Tinguirirican and Deseadan is ‘La Cantera fauna’.

on a time series of paleoclimate zone maps (Quintero *et al.* 2023; Scotese *et al.* 2021), we assigned each of our occurrences a tropical or extratropical affinity (SI Appendix, Figure S3), and analyzed genus-level diversification dynamics within each biome separately. Our findings revealed that extratropical genera exhibited diversity and diversification patterns extremely similar to those obtained from the complete dataset across the Eocene–Oligocene interval (Figure 3D-F and SI Appendix, Figure S8-11). Their diversity gradually declined from the middle Eocene to early Oligocene, with a late Eocene turnover increase (Figure 3D-F). This proximity between extratropical and continent-wide patterns was expected as extratropical taxa constitute the majority of genera present in our dataset (85.8%, SI Appendix, Table S10). Nevertheless, results are different when constraining our analyses to tropical genera (Figure 1.16 A–C). No significant change was found in the tropical origination rate, and one significant negative shift was found for the tropical extinction rate in the late Oligocene (but see later for an interpretation of this extinction peak) (Figure 1.16 A and S24). Their inferred diversity curve exhibits an opposite trend to the one obtained from extratropical genera. Tropical genera diversity increased exponentially during the Eocene and reached a plateau in the early Oligocene (Figure 1.16 C). In the late Oligocene, our reconstructions tend to suggest that tropical mammals experienced a diversity decline associated to the aforementioned increase in extinction rate. We propose this apparent decline to result from the scarcity of low-latitude fossil-bearing outcrops of Oligocene age, given that there are almost no tropical Deseadan (ca. 29.4–24.2 Ma) mammal remain unearthed to date (only two localities, CTA-32 and CTA-61 in the region of Contamana, Peru, see Antoine *et al.*

(2017)), therefore more reflecting a sampling bias rather than a true tropical SAM extinction in the late Oligocene. Consequently, after discarding this artefactual extinction peak, our results suggest that tropical lineages experienced no change in diversification rates, providing support for Wallace’s theory of tropical stability hypothesis (Wallace 1878). Also, when considering attempts to derive early Oligocene ages to east-Amazonian occurrences originally assigned to an Eocene age (Campbell *et al.* 2021), the overall tropical diversity pattern remains unchanged. Indeed, the previously highlighted Eocene increase in diversity lasts a few more million years within the early Oligocene, until the onset of the ‘decline’ – that we propose to be artificial (*SI Appendix*, Figure S33).

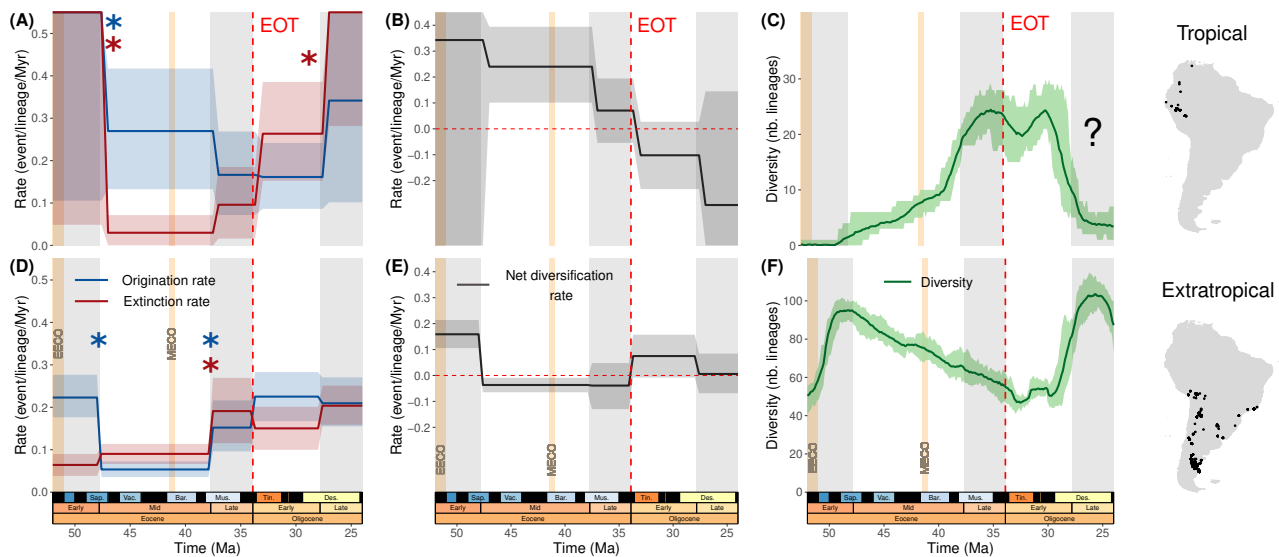


Figure 1.16: Diversification (A–B and D–E) and diversity (C and F) dynamics of South American mammal genera between the Eocene and Oligocene arranged by paleotropical affinity. Trends are assessed for the tropical (A–C) and extratropical (D–F) biomes. Origination and extinction rates are respectively depicted in blue and red. Stars indicate significant rate shifts. The timing and significance of rate shifts was assessed by the rjMCMC model (Figure S36–37). Their difference, being the net diversification rate, is represented in black. Diversity through time is plotted in green. Light brown vertical bands indicate two key climate events, the Early Eocene Climate Optimum (EECO, ~53–51 Ma) and the Middle Eocene Climate Optimum (MECO, ~41.5–41 Ma). The question mark on the tropical diversity-through-time plot (C) indicates that the late-Oligocene tropical diversity decline that is observed is likely due to the quasi-absence of fossil material of late Oligocene age (Deseadan SALMA) so far unearthed. The maps on the right show the localities of our extratropical (top) and tropical (bottom) occurrences. Geological and SALMA timescales were added thanks to the deeptime R package (Gearty 2023).

Our results suggest that highly distinct evolutionary regimes shaped tropical and extratropical mammal assemblages in South America. However, there is no clear evidence supporting commonly-assumed macroevolutionary hypotheses regarding how tropics shaped diversification processes (Figure 1.16 and *SI Appendix*, Figure S25–S29). Two macroevolutionary scenarios are usually invoked to explain the differences in diversity between tropical and extratropical biomes (Stebbins 1974). On one side, tropical lineages would have higher origination rates due to the higher diversity of habitats offered by the tropics (‘cradle’ hypothesis). On the other, tropical climate stability would make tropical lineages less

prone to extinction (‘museum’ hypothesis, *but see* Vasconcelos *et al.* (2022)). Here, tropical lineages appear to have a higher origination and a lower extinction rate than their extratropical relatives in the middle Eocene. This would typically be in line with (Rolland *et al.* 2014), who estimated temporal trends of speciation and extinction rates across latitudes using the mammal tree of life, and found that tropical mammals had a higher speciation and a lower extinction rate than their extratropical counterparts throughout the Cenozoic (*i.e.*, ‘cradle and museum’ of diversity). More recently, an emphasis was put on higher-latitude tropical extinctions to explain the build-up of present-day tetrapod diversity (Meseguer *et al.* 2022; Quintero *et al.* 2023). In alternative scenarios, varying dispersal rates can explain differences between tropical and extratropical diversity (Jablonski *et al.* 2006). There is a consensus that tropical-to-extratropical transitions are prevalent, supporting the so-called ‘out-of-the-tropics’ hypothesis (Jablonski *et al.* 2006), although extratropical-to-tropical transitions have been suggested [*e.g.*, Meseguer *et al.* (2022); Smith *et al.* (2012); Chapter 3].

Drivers behind the Eocene–Oligocene macroevolutionary dynamics of SAM

After exploring the dynamics of SAM diversity and diversification, we investigated their underlying drivers, addressing: what drove SAM diversity to decline in the 52–33.9 Ma interval? What caused their late Eocene increase in turnover? How do we explain such levels of taxonomic dissimilarity between Eocene and Oligocene SAM?

Our Multivariate Birth-Death (MBD) analyses indicate that the dissimilar Eocene and Oligocene faunas were ruled by different drivers (Figure 4). Overall, diversity-dependent effects largely impacted SAM diversification history throughout our study period, but these processes shaped both faunas differently. We recovered strong negative correlations of Eocene (post-EECO, 52–33.9 Ma) origination rate (correlation parameter $G_\lambda = -3.47$, shrinkage weight $[w_\lambda] = 0.91$) and Oligocene extinction rate ($G_\mu = -2.36$, $w_\mu = 0.83$) with self-diversity (?@fig-drivers, SI Appendix, Table S3–6). This suggests that Eocene mammal origination decreased when their diversity increased, which can be interpreted as a carrying capacity effect, that can occur at macroevolutionary scales (Storch & Okie 2019). The other diversity-dependent correlation indicates that the Oligocene mammal extinction rate decreased when their diversity increased. In the framework of the marked taxonomic turnover highlighted by our analyses, this effect might have arisen from facilitating interactions between recently-originated taxa (see forthcoming paragraph), but it could also be indirectly related to enhanced ecosystem productivity or niche partitioning events, which cannot be captured by the MBD analyses.

The MBD analyses also provided support for a positive correlation between global temperature and Eocene mammal origination rate ($G_\lambda = 0.24$, $w_\lambda = 0.84$). This explains the previously-described gradual decline in SAM diversity in the mid-late Eocene, qualitatively matching with the post-EECO overall decline in temperature (Figure 1.14 D and ?@fig-drivers B). It is now clearly established that temperature change directly impacts biodiversity, at every timescale (Condamine *et al.* 2013; IPCC 2023; Malanoski *et al.* 2024), and temperature was already identified as a key driver influencing the diversification of an endemic European artiodactyl clade at the EOT (Weppe *et al.* 2021). Based on isotope analyses of Patagonian ungulate enamel and on paleobotanical data, it has been proposed that the South American climate, at least in the Southern part of the continent, remained merely stable across the Eocene–Oligocene interval (Kohn *et al.* 2004, 2015; Pound & Salzmann 2017). However, as illustrated by Antoine *et al.* (2021), we assume that the spatial scale (*i.e.*, continental vs. global) does not affect the trends and directionality of temperature changes through time, but rather their amplitude. Hence, in the framework of our MBD analyses, the use of a global estimate of paleotemperature is still relevant, as we are more interested in trends than having actual paleotemperature values. Also,

to date (*at the time the study was conducted*), no regional paleotemperature reconstruction for the South American continent has been produced with the same resolution as the global estimate used in this study (Westerhold *et al.* 2020).

We estimated a weak but significant positive correlation between mammal extinction rate and the Andean uplift both in the Eocene ($G_\lambda = 0.005$, $w_\lambda = 0.68$) and the Oligocene ($G_\mu = 0.02$, $w_\mu = 0.83$) (?@fig-drivers, *SI Appendix*, Table S3-6). This Eocene correlation was maintained among high-latitude (extratropical) taxa when a more regionalised proxy of the pace of the Andean uplift was used (*SI Appendix*, Figure S41). Mountain-building processes are known to greatly influence species diversity (Antonelli *et al.* 2018), and previous studies illustrated the key importance of the Andes in shaping diversity patterns of a wide range of South American organisms, even in the early stages of the build-up of the chain (*e.g.*, Antonelli *et al.* 2009; Boschman & Condamine 2022; Vallejos-Garrido *et al.* 2023). In a context of active margin, the Andean orogeny has been associated with volcanic activity, which, in conjunction to subsequent forest fires, resulted in a massive dust emission. This dust emission has been postulated to promote grazer-like herbivore origination and diversification, in particular at the EOT and thereafter (Kohn *et al.* 2015; Selkin *et al.* 2015). Due to the high proportion of herbivores in our dataset (*SI Appendix*, Table S10 and Figure 1.15), we could have expected Andean uplift to be either positively related to SAM origination, negatively related to SAM extinction, or both. Here, our results suggest that Andean uplift promoted extinction among SAM faunas, thus not providing support for any of the previously-stated hypotheses. Instead, they lead us to hypothesise that the building of the Andes, possibly by reshaping the landscapes locally, re-organising climate advections, triggering volcanism or any other factor, was an obstacle, rather than a facilitator, for SAM diversification across the Eocene–Oligocene interval.

Interestingly, when working with the complete dataset, none of our proxies for land opening, *i.e.*, phytolith-derived floral composition – proxy for grassland expansion, or habitat openness (Strömberg *et al.* 2013) – and relative Leaf Area Index (rLAI) – proxy for forest density (Dunn *et al.* 2015) – were found to correlate with diversification rates. Similar conclusions were reached when restricting our analyses to notoungulates, the richest herbivorous clade of the dataset (*SI Appendix*, Figure S35). Nevertheless, fitting the MBD only to high-latitude (extratropical) taxa revealed a significant positive correlation of extinction rate with rLAI ($G_\mu = 2.56$, $w_\mu = 0.96$), and a negative correlation of origination rate with habitat openness ($G_\lambda = -0.06$, $w_\lambda = 0.62$) during the Oligocene (*SI Appendix*, Figure S40 and Table S11–S12). The interplay between grassland expansion and diversification of (herbivorous) mammal in South America has been debated in the literature (Pound & Salzmann 2017; Stebbins 1981; Strömberg *et al.* 2013). The convergent evolution of high-crowned cheek teeth and larger body sizes in several notoungulate families near the EOT, suggesting the emergence of a grazing diet, has long been regarded as evidence for grassland expansion in response to climate cooling and aridification (Stebbins 1981). However, subsequent studies have shown that grasslands did not spread as much during the mid-Eocene to Oligocene of Patagonia, while hypsodont taxa evolved and diversified (Strömberg *et al.* 2013), and more generally that the composition of South American paleobiomes remained stable during this period (Pound & Salzmann 2017). Furthermore, trait-based correlation analyses failed to recover temporal relationship between the evolution of hypsodonty and the abundance of open habitats (Solórzano & Núñez-Flores 2021). Here, our analyses provide contrasting evidence regarding the role of grassland spreading in SAM diversification. On a continental scale, no effect could be found, providing support against the common paradigm linking mammal diversification to grassland expansion. On the other hand, our result on a latitudinally-divided dataset revealed attenuated extinction and lower origination for Oligocene high-latitude grassland-dwelling taxa. It is currently difficult to determine whether this difference arises from an actual scale effect or heterogeneity in our data. Therefore, further research with increased sampling effort, especially at

low latitudes, would be needed in order to address the question of the interplay between grassland expansion and SAM diversification.

When carrying out the MBD analyses without partitioning Eocene and Oligocene faunas, we still found strong negative correlations of origination ($G_\lambda = -2.16$, $w_\lambda = 0.90$) and extinction ($G_\mu = -2.18$, $w_\mu = 0.81$) rates with self-diversity, and a weak positive correlation of extinction rate with the Andean uplift ($G_\mu = 0.003$, $w_\mu = 0.70$) (*SI Appendix*, Figure S34, Table S3–6). However, we no longer retrieved any correlation of origination rate with temperature, highlighting the prominent and exclusive role of temperature in shaping Eocene mammal diversity.

Materials and Methods

Additional methodological details are provided in the *SI Appendix* to this chapter.

Fossil data compilation and analytical framework

We downloaded and cleaned all genus-level South American terrestrial mammal occurrences from The Paleobiology Database <https://paleobiodb.org/> spanning the late Paleocene to the early Miocene (60–16 Ma) (3,384 occurrences; *SI Appendix*, *Extended Materials and Methods* and Tab. S1). Cleaning consisted of a manual revision of the taxonomic name (*e.g.*, synonymies, missing open nomenclature elements, typos) and time range of each occurrence (*SI Appendix*, Figure S4 and *Manual Cleaning Details*). SAM fossil data are prone to a large spatial heterogeneity, with most fossil localities located close to or high in the Andes – in their current conformation (*SI Appendix*, Figure S3). Using palaeo-coordinate reconstructions (Jones *et al.* 2023; Scotese & Wright 2018) and paleobiome maps (Scotese *et al.* 2021), we assigned each occurrence a tropical or extratropical affinity, and collapsed their associated dental ecomorphotype-derived trophic classes into four categories: carnivorous, herbivorous, omnivorous and insectivorous (*SI Appendix*, Tab. S10).

We analyzed our database using PyRate 3 (Silvestro *et al.* 2019) and approximated the posterior distribution of the origination and extinction times (T_s and T_e , respectively) of each genus and the rate of the preservation process using the rjMCMC algorithm. We then used these T_s and T_e to estimate the underlying diversification process (origination and extinction rates) with the Birth-Death model with Constrained Shifts (BDCS) (?) (*SI Appendix*, *Extended Materials and Methods*).

We ran the rjMCMC model for 50 million generations and sampled every 50,000, with a time-dependent preservation process allowing for rate shift between geological stages (*SI Appendix*, Figure S5–6 and *Extended Materials and Methods*). We allowed for a gamma-distributed preservation heterogeneity across lineages. We ran the BDCS model for 10 million generations, sampled every 10,000, and allowed diversification (*i.e.*, origination and extinction) rates to vary between each geological sub-Epochs (*SI Appendix*, Figure S1). Diversification analyses were replicated 20 times after randomly drawing fossil ages between their upper and lower age boundaries, and posterior estimates across all replicates that achieved convergence ($n = 11$, *SI Appendix*, Figure S42) were combined. To test the robustness of our macroevolutionary inference, we conducted sensitivity analyses by either removing singletons or subsampling our fossil database to unique locality-lineage combinations (*SI Appendix*, *Extended Materials and Methods*). We assessed the convergence of each parameter with the pymc 5.9.2 python library (Wiecki *et al.* 2023). We considered parameters achieved convergence if their Effective Sample Size (ESS) was above 200.

Age-dependent Extinction (ADE) model

We assessed the effect of taxon age on its extinction probability by fitting the ADE model to our genus-level occurrence data (Hagen *et al.* 2018). We ran the model for 100 million generations and sampled every 50,000, with preservation rates varying between geological stages and across lineages according to a Gamma model. We focused on the shape (Φ) of the Weibull distribution, knowing that $\Phi < 1$ indicates a decreasing extinction probability with increasing taxon age and $\Phi > 1$ indicates an increasing extinction probability with increasing taxon age (no effect if $\Phi = 1$). Φ was considered as significantly different from one if one was excluded from a significant proportion of its posterior distribution. We performed 20 independent ADE analyses of randomised datasets before and after the increase in turnover to assess the age selectivity of this taxonomic renewal (*SI Appendix*, Figure S8-11, S43 and *Extended Materials and Methods*). We monitored parameter convergence with the pymc 5.9.2 python library, and parameters were considered convergent if their ESS was above 200.

Selection of environmental variables

To investigate possible environmental correlates for the changes in SAM diversification dynamics over the Eocene–Oligocene interval, we selected two global (*i.e.*, atmospheric temperature and sea level) and four continental-scale (*i.e.*, Andean uplift, Neotropical plant diversity, forest density and habitat openness) variables (*SI Appendix*, Figure S40 and Tab. S2). Temperature is known for influencing macroevolutionary processes across several groups, including mammals (Barnosky 2001; Condamine *et al.* 2019a), and sea level can impact the connectivity of habitats. Even in its early phase, the Andean uplift was evidenced as a major diversification driver for many South American groups (Boschman & Condamine 2022). Diversity of SAM, in particular herbivorous taxa, can be controlled by plant diversity. Finally, relative Leaf Area Index (proxy for forest density, Pound & Salzmann (2017)) and patterns of relative abundance of open habitats (Palazzesi & Barreda 2012; Solórzano & Núñez-Flores 2021; Strömberg *et al.* 2013) are both indicators of the relative abundance of grasslands, which has long been regarded as shaping herbivore diversity (Stebbins 1981). As highlighted by Weppe *et al.* (2023), it is important to consider regional-scale variables because environmental changes across the Eocene–Oligocene interval were shown to be regionally heterogeneous (Pound & Salzmann 2017; Toumoulin *et al.* 2022).

Multivariate Birth-Death (MBD) model

We carried out environment-dependent diversification analyses involving the aforementioned variables by running the MBD model included in PyRate (Lehtonen *et al.* 2017). This model carries out two independent correlations involving a set of environmental covariates and diversification rates (λ and μ) newly-estimated from empirical T_s and T_e . Each variable is assigned to two correlation coefficients, one for the correlation with λ ($G_{i,\lambda}$, where i is the index of the variable in the set of covariates) and the other for the correlation with μ ($G_{i,\mu}$). An MCMC algorithm jointly estimates baseline origination and extinction rates and all the G and G_{μ} , while controlling for over-parameterization. Each G_{λ} and G_{μ} is assigned a shrinkage weight ($w(G_{\lambda})$ or $w(G_{\mu})$), ranging between 0 and 1, testing the significance of the correlation. A correlation of coefficient $G_{i,\lambda}$ is considered significant if $w(G_{\lambda,i}) > 0.5$ and if a sufficiently high proportion of its posterior density does not overlap with 0. Each set of MBD analyses was run across 10 million iterations, sampling every 10,000. We carried out exponential correlations. As we expected Eocene and Oligocene faunas to be influenced by different drivers, we also performed

independent MBD analyses for the Eocene (post-EECO, *i.e.*, 52–33.9 Ma) and for the Oligocene (33.9–23.03 Ma) taxa (*SI Appendix*, Figure S44). Parameter convergence (*i.e.*, $ESS > 200$) was assessed using pymc.

Data, Materials, and Software Availability

All the occurrence data used in this study are available on a Figshare repository (<https://doi.org/10.6084/m9.figshare.26495989>) and all the data analysis and visualization codes have been deposited in GitHub (https://github.com/Buffan3369/Neotropical_mammals.git and <https://github.com/Alexis-Marion/CorsaiR.git>).

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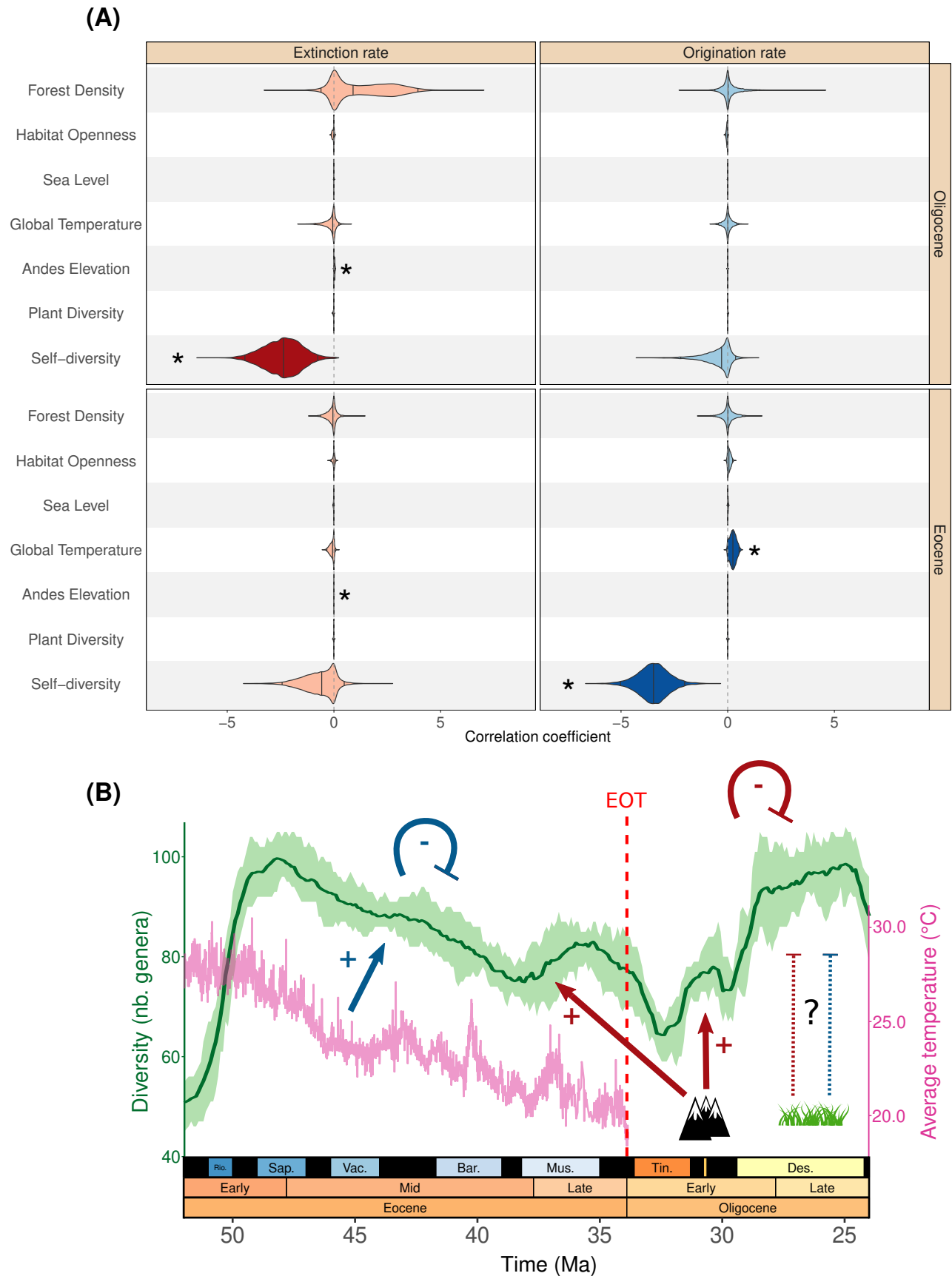


Figure 1.17: Bayesian estimation of correlation parameters on origination (blue) and extinction (red) rates with paleoenvironmental variables. **(A)** We analyzed separately the post-Early Eocene Climate Optimum (52–33.9 Ma, bottom row) and Oligocene (33.9–23.04 Ma, top row) mammal faunas, therefore not treating our dataset as a whole throughout our study period. The 2.5%, 50% and 97.5% quantiles are indicated in the distribution of each

Supporting Information Appendix for Chapter 1

Supplementary text

1. What is a mass extinction?

The expression “mass extinction” has been historically introduced to characterize the punctual-in-time dramatic diversity drop-downs that could be drawn from the first global-scale compilations of the marine fossil record Sepkoski *et al.* (1981). It is now admitted that, given a taxonomic group, a mass extinction event can be defined as a global loss of the majority of the lineages belonging to this group, occurring within a short time period Raup & Sepkoski (1982). The time frame of a mass extinction is always considered as geologically instantaneous, meaning that it occurs on a timescale ($\approx 10^3 - 10^5$ years) below the resolution of the records used to study them (*i.e.*, fossils, $\approx 10^6$ years) (?).

2. Extended Materials and methods

Fossil data compilation

We first downloaded all Cenozoic mammal occurrences at least with a genus-level resolution from South America from The Paleobiology Database (?base_name=Mammalia&taxo_reso=genus&cc=SOA) on 2023-12-11 (n=10,440 occurrences). Because diversification regimes do not occur in the same way on land and in the sea (?), and because the scope of this paper is terrestrial, we removed taxa with marine affinities, *i.e.*, occurrences assigned to the orders ‘Cetacea’ and ‘Sirenia’ or the families ‘Otariidae’ and ‘Phocidae’ (n=354 occurrences). Subsequently, as ref. Tarquini *et al.* (2022) performed an intensive cleaning of a significant proportion of South American mammalian fossil occurrences (especially Sparassodonts and Native Ungulates), we matched the time ranges of the occurrences we shared with them to those that they proposed (n=2,998 occurrences). For the remaining occurrences, we ensured that their time ranges matched the boundaries of their associated interval. For instance, the Mustersan South American Land Mammal Age (SALMA) occurrences were all found to be associated with the 48–42 Ma interval. However, radiochronological evidence has shown that this age is younger, constraining its boundaries to 38.2–35 Ma (conservative estimates have even constrained it to 38.2–38 Ma, based on the only clear datations from well-defined levels) Dunn *et al.* (2013) (?@fig-s1). It is quite concerning to see that such major changes remain uncorrected, especially in public encyclopedias, often used as primary sources of information for non-specialists (*e.g.*, <https://en.wikipedia.org/wiki/Mustersan>). Age boundaries were expressed as either an International Chronostratigraphic Chart (ICC) stage/epoch or a SALMA. In the latter case, we used a SALMA scale built de-novo based on a literature review. Then, based on the literature and the expertise of the co-authors, we manually revised the entire dataset, occurrence by occurrence. This ‘manual cleaning’ aimed at correcting errors in the database, mostly regarding the age range and the taxonomic assignment (*e.g.*, synonymized genus names, typos, missing elements of open nomenclature) of each occurrence (?@fig-s2). We

intended to make this step as clear as possible by documenting most of the changes we made, and why (see *Manual Occurrence Cleaning*). Finally, in order to decrease the number of parameters to estimate, we restricted our dataset to the late Paleocene-Early Miocene (60–16 Ma). The final dataset comprised 3,384 occurrences, including 529 genera and 1,108 identified species (see ?@tbl-s1 for a more detailed description of the data). Occurrence data can be found at <https://doi.org/10.6084/m9.figshare.26495989>.

Further input treatments

Subsequent data manipulation and visualization were carried out using the Tidyverse (?) suite of R packages, the paleoverse (?) and the deeptime (?) R packages. Out of 3,384 occurrences, 2,108 occurrences (representing *approx.* 62% of the total) were assigned to a SALMA. Due to temporal gaps in the South American Mammal fossil record this timescale is not continuous (?@fig-s1). Yet, gaps can highly bias fossil-based diversification inferences, as, for instance, shown in insects (?). Therefore, we decided to assign sub-epoch ages to the SALMA-associated occurrences (?@fig-s1; -?@fig-s2), ensuring ourselves to sample the whole study period more evenly. As illustrated by many studies (Condamine *et al.* 2019b; Tarquini *et al.* 2022), the taxonomic scale impacts fossil-derived diversification patterns. Defining species in the fossil record is difficult and prone to debate, and working at the species scale is particularly sensitive to biases such as synonymy or missing open literature elements (*e.g.*, ‘*cf.*’, ‘*aff.*’, etc.). Therefore, we analyzed our mammalian fossil database both at the genus and species levels (?@fig-s2), but privileged the genus scale for our macroevolutionary conclusions. We further applied several treatments to our data. First, we tested the effect of removing singletons because our genus-level and species-level occurrence datasets comprise 36% and 54% of singletons, respectively (*i.e.*, taxa only represented by a single occurrence in the entire dataset; ?@tbl-s1). It is worth noting that including singletons in diversification inferences is still under debate (Foote 2000; Guinot & Condamine 2023; ?). Also, PyRate was validated using simulated data with a proportion of singletons that, to our knowledge, never exceeded 34% (Silvestro *et al.* 2019). Hence, we possibly ignore how the model might behave with such high proportions of singletons as those we have, though it was found to be robust against data incompleteness. Second, we did not allow a given genus (or species) to have geographically and temporally-identical occurrences (as done by (?)). Next, in order to have a better view on the different tempos of diversification between tropical and extratropical biomes, we assigned each occurrence a tropical or extratropical affinity. Traditionally – both when dealing with extant or fossil lineages – such a geographic splitting is made by defining a latitudinal threshold above (or under, depending on the hemisphere) which occurrences are considered tropical. The latter is commonly set as -23.4° in latitude, but sometimes also -23° , -23.5° or even -30° . However, such a threshold is unsatisfactory, mainly because it does not consider tropics’ expansions and contractions through geological times (see refs. Quintero *et al.* (2023) for a discussion on the topic). Therefore, based on their middle age, we binned our occurrences into 5-Myr time bins and reconstructed their paleocoordinates (using the mid age of the bin as reconstruction age) with the PALEOMAP Global Plate Model Scotese & Wright (2018). Paleorotations were produced via an access to the GPlates Web Service (<https://gwsdoc.gplates.org/>) API implemented in the `paleorotate` function of the paleoverse R package (Jones *et al.* 2023). Then, to assign them a tropical/extratropical affinity, we overlaid these point data with Köppen-Geiger paleoclimate zones produced every 5 My (Quintero *et al.* 2023; Scotese *et al.* 2021). Finally, our fossil occurrence data came with ‘diet’ assignments, derived from dental ecomorphotypes, which we further revised and collapsed to only keep four different classes: carnivorous, herbivorous, insectivorous, and omnivorous.

Bayesian inference of diversification accounting for preservation

We used the Bayesian framework of fossil-based diversity and diversification inference accounting for preservation implemented in PyRate v.3 (Silvestro *et al.* 2014b; Silvestro *et al.* 2014a). Relying on birth-death and preservation processes, PyRate jointly estimates the times of origination and extinction and the preservation rate of each taxon, as well as the origination and extinction rates of the underlying birth-death process. We approximated the posterior distribution of each parameter through Reversible-Jump Markov Chain Monte Carlo (RJMCMC) Silvestro *et al.* (2019). The RJMCMC algorithm provides an objective estimate of the number of diversification rate shifts, but empirical case studies found that it could also be sensitive to gaps in the fossil record (?). Hence, we also analyzed our fossil dataset under a Birth-Death model with Constrained rate Shifts (BDCS) (?). In the BDCS model, origination and extinction rates can only vary in a piecewise-constant way between a predefined set of time bins. Given the temporal heterogeneity of our data (?@fig-s1 and -?@fig-s4), this framework was particularly relevant to us because the BDCS model can smooth rates through time when rates are very different within a bin due to preservation issues. We both tested geological stages and 5 My-long intervals as input bins. To achieve convergence, when working with full datasets, we ran 50,000,000 iterations at the genus level and 100,000,000 iterations at the species level, and sampled every 50,000. When working with taxonomically- or diet-specific datasets, we ran 20,000,000 iterations, sampling every 20,000. For each tested condition, we ran PyRate on 10 or 20 datasets of independently-sampled ages and retained those of which the majority of the parameters (posterior above all) converged. Parameters were considered convergent when their Effective Sample Size (ESS) was above 200. Convergence checks were assessed thanks to the PyMC 5.9.2 python library Wiecki *et al.* (2023). For each set of runs, burn-in value was specified in an automatized pipeline as the value maximizing the ESS of the parameters (choosing between 1%, 5%, 10%, 25% and 50% of the posterior distribution), rather than using the same fixed value every time. Parallel computing was made possible thanks to the parallel software (?). In the RJMCMC analyses, we allowed for a minimal time window between two origination/extinction rate shifts of 1 My (`-mindt 1`). We also allowed for preservation rate heterogeneity across taxa (`-mG`) as it is more realistic to consider uneven preservations across lineages (some lineages undergo better preservation than others). We set Gamma priors of shapes 1.1 (default value) and model-estimated rates for both origination and extinction rates (`-pL 1.1 0` and `-pM 1.1 0` options). The preservation rate prior was also sampled in a Gamma distribution of model-estimated rate, but we put a slightly larger shape (`-pP 2 0`), offering more flexibility Guinot & Condamine (2023). We assessed the most suitable preservation model for our data using the maximum-likelihood based model selection test included in PyRate (`-PPmodeltest` option), which resulted in favor of the Time-dependent Preservation Process (TPP) model allowing rate shifts between every geological stage (?@fig-s5; -?@fig-s6). Last, we tested for the sensitivity of our analyses to singleton removal (`-singleton 1` option) (?@fig-s12; -?@fig-s13). As the BDCS model takes longer to run than the RJMCMC model, we ran the former directly based on the origination and extinction times (respectively T_s and T_e) table estimated with the latter (`-d` option). In fact, the joint estimation of T_s , T_e and the preservation process is the same between both models. What changes is the diversification (birth-death) model they fit to the reconstructed diversity to assess the origination and extinction rates through time (λ and μ , respectively). BDCS models were run across 10,000,000 iterations, sampling every 10,000. All subsequent analyses were performed in R v4.4.0 (?).

Chapter 2. A matter of size: what underpinned the rise and demise of the largest rodents that ever lived?

Chapter 3. Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Discussion

EOT: A mass extinction? Could this, as proposed by Bambach (2006), reflect the fact that the definition of a mass extinction should incorporate degrees of freedom?

Limitations

Lack of time-continuous proxy for precipitations.

Lack of well-constrained phylogenetic framework, as many of the endemic Cenozoic South American mammal fauna has gone extinct

Traits mediate biotic interactions

Terminology, definitions and other macroevolutionary enigmas

Defining a species?

Inferring diversity/diversification patterns from the fossil record: Signor & Lipps effect Signor & Lipps (1982)

Perspectives

An almost closed system: founding migrations in the Palaeogene history of South American mammals

Hystricognath rodents & Platyrrhine primates. Introduce the debate. We lack fossil evidence for the earliest comers, oldest occurrences in Western Amazonia whereas most likely come from Afro-Asian ancestors, so should have arrived by the East. Polyphyletic/several crossing events? Summarised in Marivaux *et al.* (2023)

A recent quantitative study suggested that the diversification of rodents and primates negatively affected the diversity of the endemic groups (Pino *et al.* 2025).

There is still some work to do to accurately elucidate continent-scale macroevolutionary dynamics of Caviomorph rodents (others than Chinchilloids, a contribution of this thesis): set the scene for GIGARAT

Sustainable late Cenozoic connection to North America: what drove the asymmetry of the GABI?

Set the scene for IMBALANCE

How about neutral theory?

Future directions towards an improved understanding of the emergence and dynamics of South American ecosystems

DIACHRONANDES, PHENOMENA

Concluding remarks

Glossary

This section intends to provide simple definitions to key notions and concepts to which this thesis refers. Note that some are complex and could be the subject of entire research works. Thus, these definitions do not aim to be as exhaustive as possible, but rather, to improve the reader's understanding of this manuscript.

Adaptive radiation. Rapid *diversification* of a group taking advantage of a previously unexploited ecological dimension of the environment, generally spurred by a *key innovation*. Adaptive radiations are accompanied with ecological divergence. This concept has been proposed by Simpson (1944).

Biogeography. Discipline aiming to describe and understand the spatial distribution of species.

Clade. Given a multi-species sample, a clade is an exhaustive subsample of all species sharing a common ancestry. Clades are referred to as *monophyletic*.

Diversification. The diversification process summarises the net evolutionary changes in a *clade* across time, that is, the gains (modelled by the *speciation* process) minus the losses (modelled by the *extinction* process). As such, it represents the process generating clade's diversity. The intensity of the diversification process (called diversification rate, often noted $d(t)$ for a given time t) is thus the difference between the rates of speciation and extinction, leading to $d(t) = \lambda(t) - \mu(t)$.

Ecological niche. The concept of ecological niche may be understood in different ways. In macroecology, the niche is more often viewed from a perspective of the species' requirements, therefore following the framework proposed by Grinnell (1917) and later Hutchinson (1957). Considering a landscape made up with n ecological variables (*e.g.*, climate, topography, abundance of other species, abundance of trophic resources, ...), the ecological niche of a given species theoretically represents the portion of the landscape where the species occurs. As such, it can be viewed as an n -hyperpolygon encompassing the species' tolerance range following each variable (*e.g.*, 15–25°C, 0–1000m, ...). Other definitions exist, such as the Eltonian niche Elton (1927), more centered on species' impact on the environment.

Extinction. Process of disappearance of all representatives of a given lineage. In macroevolutionary modelling, extinction rates (expressed in event per unit of time) are often noted $\mu(t)$.

Latitudinal Biodiversity Gradient (LBG). Pattern of increasing species diversity with decreasing latitude (*i.e.*, within a given unit of the globe's surface, the number of species within it increases as we get closer to the equator), found in a wide range of group.

Mass Extinction. Geologically-instantaneous event (order of magnitude $\approx 10^5 y$ Guinot & Condamine (2023); Li *et al.* (2026)) characterised by the loss of a significant proportion of the biosphere. Five major mass extinctions have been acknowledged to have impacted the Phanerozoic history of life, termed the “Big Five mass extinctions” (Raup & Sepkoski 1982), but the definition of mass extinctions has been proposed to be extended to other intervals (Bambach 2006; Hoyal Cuthill *et al.* 2020).

Stochastic process. As opposed to *deterministic processes*, always resulting in the same output (*e.g.*, Ordinary Differential Equations in Physics), *stochastic processes* are built upon one or several *random variables*, thus inserting some degree of randomness in their outputs governed by known probability

laws. Stochastic processes are thus often appropriate to model biological systems, having some part of unpredictable behaviour (often termed ‘noise’) that cannot be rendered by deterministic processes.

Speciation. Origination of a new species from an ancestral lineage. There are several known types of speciation. Speciation is called allopatric when a physical barrier (*e.g.*, a mountain, a river) subdivides an ancestral population into several sub-populations, thus interrupting gene flow and driving to the split in as many species as initial subsets by divergence accumulation. Conversely, a the barrier to gene flow arises within a population where individuals remain in connexion (*e.g.*, behavioural barrier), people refer to *sympatric* speciation. When speciation arises from a founder effect, *i.e.*, colonisation of a new area (*e.g.*, an island), one refers to *peripatry*. A particular case of peripatry, when the colonised area is adjacent to the range of the initial species, is called *parapatry*. Given these several modes, modelling speciation can be done in different ways, defined in this glossary (*cladogenesis*, *anagenesis*, *budding*,). In macroevolutionary modelling, speciation rates through time, expressed in number of events per unit of time, are often noted $\lambda(t)$

Trait.

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